Green Chemistry

ROYAL SOCIETY OF CHEMISTRY

CRITICAL REVIEW

Published on 06 January 2020
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Cite this: *Green Chem.*, 2020, **22**, 612

Greener synthesis of lignin nanoparticles and their applications

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Received 11th August 2019,
Accepted 6th January 2020

DOI: 10.1039/c9gc02835h

rsc.li/greenchem

The remarkable absorption capacity, biodegradability and non-toxicity of lignin nanoparticles enable them to be appropriate vehicles for drug molecules and inorganic particles. Lignin nanoparticles have potential applications in the biomedical and environmental fields, such as tissue engineering or regeneration, artificial muscles, and strong underwater antifouling materials. Additionally, the synthesis of nanoparticles using physicochemical methods can generate toxic and hazardous effects in the environment and often need expensive materials and instruments. Thus, applications of cost-effective, safer, simple, sustainable and eco-friendly approaches for fabricating nanoparticles which comply with the "green chemistry" principles are very important. In this review, significant recent advancements pertaining to the biomedical and therapeutic applications of lignin nanoparticles are highlighted including their greener synthetic options.

1. Introduction

Plant biomass includes lignocellulosic materials, carbohydrates, lignin, proteins, fats, vitamins, dyes, flavors, and aromatic essences of various chemical structure-types.[1–10] Lignocellulosic materials, with an abundance of about 70% of the total plant biomass,[11] are complex materials produced by plants during cell wall formation, and have three main components namely micro-fibrils of cellulose, hemicellulose and lignin.[12] The concept of green technology biorefinery is to achieve complete utilization of most lignocellulosic biomass including lignin components *via* greener means.[13]

Lignin has been essentially produced and used by the pulp industry, but nowadays it can be explored as a building element in the design of bio-based materials (lignin-based materials epoxy or polyurethane), a UV stabilizer, an antioxidant and a low- or high-value additive for polymers (Fig. 1).[14–16] An important source of lignin is the black liquor obtained from wood, corn stalks, straw, bagasse, and the like, in processes that entail the isolation of cellulose fibers or pulp from lignin.[17] Lignin is a complex and irregular biopolymer containing randomly cross-linked phenylpropanoid units (coumaryl, coniferyl, and sinapyl alcohols) and is located in plant secondary cell walls. Based on these monolignol units, the lignin building blocks, *p*-hydroxyphenyl, guaiacyl and syringyl units, are generated which are connected *via* several types of linkages, mainly ether bonds such as aryl- or phenyl ether and carbon–carbon bonds such as biphenyls, diphenyl ethane or pinoresinol.[16,18,19]

Among different types of biopolymers, lignin has one of the complex structures and, not surprisingly, exhibits various properties namely insolubility in an aqueous environment and its enzyme-like activity can be harshly abridged if not abolished in organic solvents. Lignin is associated with other components such as cellulose and hemicellulose in a given plant/carbohydrate essentially comprising an aromatic structure as the building block.[20] Indeed, various methods have been used for the quantitative estimation of lignin and are classified as direct (its separation and weight)/indirect methods that gauge the functional/characteristic groups of lignin, for example methoxy groups are determined and with a suitable multiplication factor, the percentage of lignin in a given plant is estimated;[21–24] the variation is observed with the source of lignin and the technology deployed in its isolation. Depending on the processing method or technology used, lignin chemistry relies on the underlying reactions such as acid-catalyzed,[25–27] base-catalyzed,[28,29] reductive,[30–32] oxidative,[33,34] and thermal reactions;[35–37] its chemistry including fractionation, depolymerization, and upgrading of lignin has been elegantly summarized.[38]

Lignin has more complexities when compared to other constituents because of its structure and the associated functional groups which can be acetylated and alkylated as per the need for the desired outcome in addition to several other reactions such as halogenation, reduction, nitration, sulfonation, oxi-

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dation, catalytic hydrogenation under pressure, hydrolysis, and fusion with alkalis or metals. Lignin is generally depolymerised to develop newer monomers[39] but several complexities exist as it is strongly held together by some of the strongest bonds namely C–C and C–O and further structural features are affected by the methods used in its extraction. These methods include the Björkman process,[40] cellulolytic enzyme lignin,[41,42] enzymatic mild acidolysis lignin,[41,42] ionic liquid treatment,[20,43] and the organosolv process.[39,44,45] Furthermore, the starting materials also affect the structures, properties and chemistry of lignin.[46] Different chemistries and catalysis involved in lignin valorization have been reviewed which highlight the importance and underlying complexities of the lignin structure.[38]

Lignin has constantly been treated as waste and applied in low-value applications including heat and electricity production. But, its abundance in nature could possibly elucidate the problematic issue of the promptly diminishing resources if it is positively translated into a renewable resource or valorized to higher value materials.[47,48] In this critical review, greener syntheses of lignin NPs and their significant recent applications are discussed.

2. Isolation and purification of lignin

Isolation and purification of lignin can be accomplished from various sources of lignin such as woody biomass, trees, wheat straw, prairie cordgrass, switch grass, and corn stover. In the procedure of cellulose extraction, lignin is obtained as a by-product in most industries, and is typically burned or discarded in a waste stream, but with increasing attention to lignin in the biorefinery industry, different approaches have been developed for the direct extraction of lignin, each with its own advantages and disadvantages.[17] Indeed, the extraction methods can cause partial cleavage in the lignin bonds, which affects the functional groups and molecular weight of the isolated structure; biomass pretreatments include chemical, physical, mechanical, solvent fractionation and biological pulping processes.[49] These processes have their advantages and disadvantages; for example, while the pretreatment of wheat straw using ionic liquids addressed some disadvantages of traditional approaches in terms of equipment corrosion, high temperature, and pressure, their applications on the industrial scale were restricted by their higher synthesis cost and complicated synthetic procedures.[50]

2.1. Isolation

2.1.1. Mechanical pulping procedure. The mechanical process is the oldest pulping method, which uses mechanical energy to weaken and release cellulose fiber from the biomass. Logs are first de-barked and then ground down with large abrasive grinding wheels and the resulting mixture is suspended in water to form a pulp.[51] Although yields from mechanical pulping contain a large amount of lignin, the



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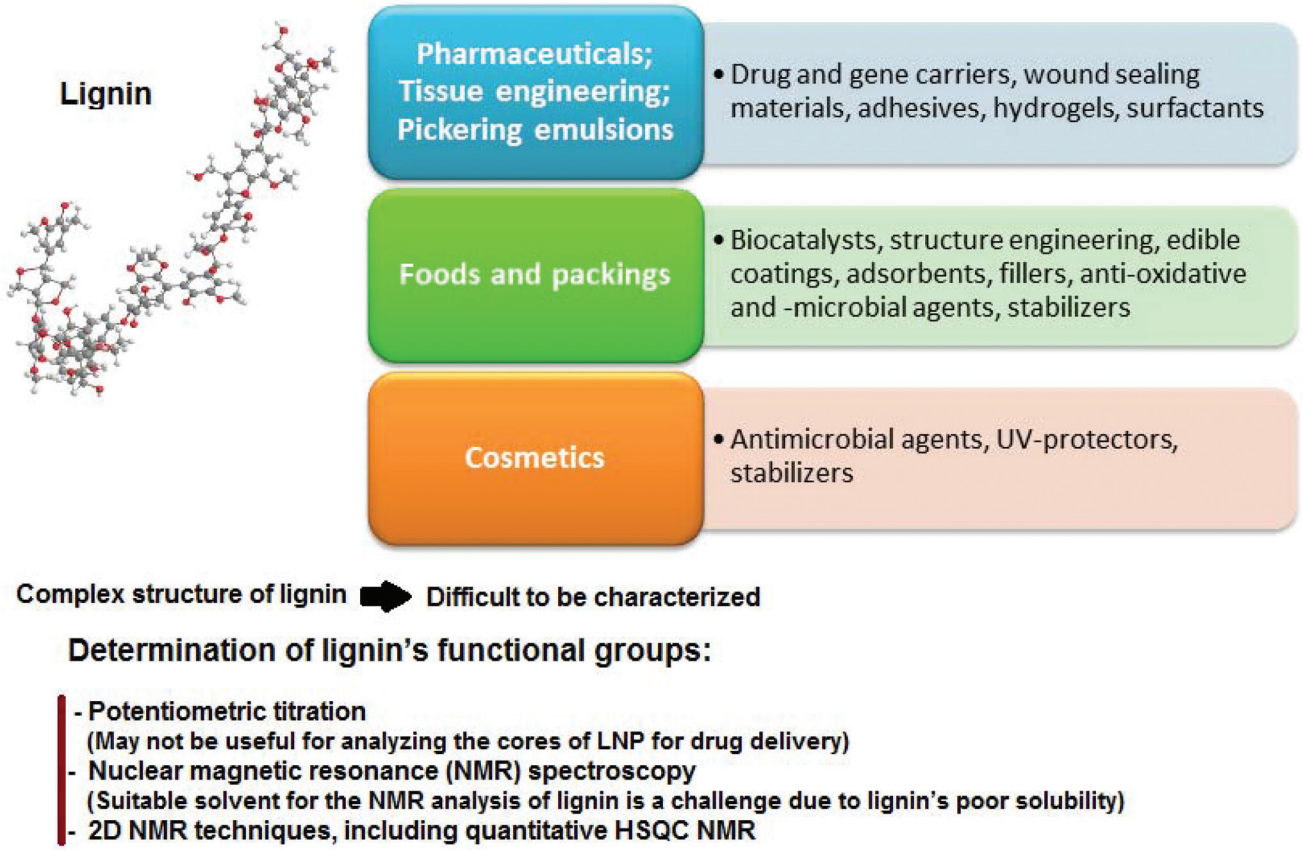


Fig. 1 Lignin as one of the most complex biopolymers.

resulting fibers are much smaller in this method compared with those from other chemical methods. Also, during mechanical pulping processes, lignin is not recovered.[52]

2.1.2. Chemical procedures. In most industries, chemical processes are desired; these procedures include acidic and alkaline pretreatments. Depending on the pulping additive, these can also give rise to a sulfur-containing lignin. This category of pretreatment which is the main commercial extraction process includes Kraft and lignosulfonate lignins.[47,53,54]

2.1.2.1 Lignosulfonate process. The lignosulfonate (sulfite) process, the oldest acid pulping approach, is initially introduced for the extraction of cellulose in a paper production process. This method is based on the application of sulfite and bisulfite in combination with several counterions. The pulping liquor is prepared by sulfur combustion under a controlled oxygen atmosphere, and the resulting sulfur dioxide is reacted with water, giving rise to sulfuric acid. The counterion source is added to the solution as the corresponding carbonate or hydroxide salt. During this method, wood is digested in sulfurous acid (H_2SO_3). The pH of the process can be changed from 1.5 to 5, based on the amount of the base introduced. The temperature can also be kept between 120 and 180 °C for 1–5 hours.[55] The isolated polymer contains 5% sulfur in the form of sulfonate groups that is highly cross-linked, soluble in water and includes carbohydrate fractions as impurities. The reaction mechanism minimizes the cleavage of side chains. Indeed, the treatment favors the cleavage of α-O-4 linkages rather than β-O-4 linkages. Additionally, the occurrence of condensation events through reaction of the intermediate carbocation with an electron-rich carbon atom leads to a lignosulfonate with high molecular weight and dispersity index. This pulping method involves a complex isolation procedure for the water-soluble lignosulfonate: initially a complexation with a long-chain alkylamine forms a water-insoluble product, then a subsequent organic extraction of the complex, and finally regeneration of lignosulfonate by addition of a base.[55,56]

2.1.2.2 Kraft (sulfate) procedure. The Kraft process is an alkaline protocol which is mostly deployed in the paper industry and accounts for about 90% of the total production of pulp worldwide.[56] In the Kraft approach, white liquor such as NaOH and sodium sulfide (NaS_2) both act as nucleophiles, breaking lignin–carbohydrate linkages and giving rise to chemically resistant isolated cellulose. The process occurs at a temperature of 175 °C in aqueous solution for 2 hours, and at the end production of dark liquor ensues, comprising a mixture of inorganic pulping chemicals, including sodium sulfide and sodium hydroxide with small amounts of sodium/calcium carbonate, sodium thiosulfate, hemicellulose, and

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lignin. Unlike the lignosulfonate (sulfite) procedure, in this approach both α- and β ether linkages can undergo cleavage. In comparison with lignosulfonate, Kraft lignin has a high purity as it contains less sulfur, fewer carbohydrate residues, and inorganic impurities. Also similar to lignosulfonate lignin, Kraft lignin is mostly used to power paper mills. Indeed, the sulfur content in both processes diminished the number of lignin applications.[57]

2.1.2.3 Soda and alkaline procedures. Alkaline and acidic processes are applied for the majority of cellulose and lignin production in the world because of their high energy efficiency. Soda pulping is one of the simplest forms of an alkaline process. This process is generally applied for non-woody biomass such as straw, sugarcane, and bagasse. In this method, lignocellulose is treated with NaOH solution (10–15 wt%) at a temperature of around 150–200 °C. Delignification is a consequence of the saponification of ester bonds between lignin and hemicellulose, while partial lignin deconstruction ensues *via* the cleavage of the α- and β-ether linkages. Soda lignin lacks any additives such as sulfur. Thus, lignins derived from the soda process may be more suitable and have added advantages for high-value applications such as in composites and bioplastics.[58]

2.1.2.4 Organosolv procedure. The organosolv process is a wood-pulping method which uses organic solvents to dissolve away lignin and hemicellulose from lignocellulose biomass; they are based on the hydrothermal treatment of biomass with a mixture of water, an organic solvent (methanol, ethanol, acetone, organic acids, amines, and ketones), and occasionally additional additives to produce a high-quality sulfur-free lignin. The ensuing lignin demonstrates the small number of changes in this method compared to the native lignin.[59] The organosolv treatment cleaves the carbohydrate–lignin bonds. The resulting lignin from the organosolv process is purer than that from other methods but it is too expensive. Consequently, the yield from this process does not have a high molecular weight; for this reason, OS-lignin is a good candidate for the preparation of polymers, such as bio-based phenol-formaldehyde resins, polyurethanes, and polyesters. At the end of the process lignin is recovered by addition of a large amount of water. The addition of an acid catalyst to the pulping media causes the cleavage of more ether-linkages, but also favors the occurrence of intramolecular condensation reactions, giving rise to a more complex lignin structure.[53,60] The major advantage of this strategy is the recovery of lignin by the simple concentration of the organic phase and the use of a biomass-derived solvent.[61]

2.2. Lignin purification

Lignin purification is a separation of lignin from cellulose and hemicellulose or other organic components that are found in the biomass. However, the purity of lignin can also refer to the metal-salt or ash content of lignin. Lignin purification is an important step in evaluating these materials. Some investigations have revealed that the primary sodium content of Kraft lignin is too high for some high-value applications. For instance, the presence of inorganics when spinning carbon fibers causes defects during carbonization, thus weakening the fibers.[56,62,63] In one study, high purity lignin has been isolated by eliminating cellulose and hemicellulose *via* a nanofiltration process after the ultrafiltration of black liquor which can lead to purer products.[64] Researchers have accelerated the efficacy of ash removal from Kraft black liquor lignins *via* a step by step reduction of pH, and then collection of fractions at pH intervals of 0.5.[65]

3. Eco-friendly and sustainable synthesis of lignin NPs

Typically, the production of nanomaterials using physico-chemical methods can generate toxic effects in the environment and often needs expensive materials. Therefore, application of cost-effective, safer, simple and eco-friendly methods for synthesizing nanomaterials that comply with the green chemistry principles is vital. Green chemistry motivates us to apply renewable resource-based feedstocks to produce various nanomaterials; generation of nanomaterials *via* greener bio-based methods or applying renewable resource feedstocks is an ecological, innovative and promising endeavor.[66–68] NPs from renewable resources are biodegradable in nature and show potential to be applied in various sustainable agricultural applications. Lignin NPs have been prepared by various approaches, including CO_2 saturation, solvent exchange of pure lignin, acetylated lignin and dialysis, flash precipitation, sonication, water-in-oil microemulsion techniques, and precipitation methods with pH lowering with an acid.[66–68] Indeed, there has been success in the production of lignin NPs *via* different protocols for various applications;[69–72] most recent advances have been targeted towards biotechnological and nanotechnology applications.[71,71] NP production from lignin is an eco-friendly option and in accordance with the green chemistry principles, it utilizes abundant lignin available in the form of industrial/agricultural waste. In this section, we describe several greener methods for the assembly of lignin NPs with inherent antioxidant, UV-absorbing and antimicrobial properties which can be applied as reinforcing agents in nanocomposites, in drug delivery and as gene delivery vehicles for biomedical applications (Table 1).

The conversion of micro-lignin to lignin NPs is promising towards attaining sustainability, because lignin prepared from industrial and agricultural waste, and *via* green chemistry principles, stimulates their applications. For instance, the high yield production of lignin NPs (15–20 nm, and spherical) has been reported in various media, including castor oil, ethylene glycol and water, and their effects on sizes and morphologies have been evaluated. The production of lignin NPs in this study, covered nine out of twelve green chemistry principles; synthesis was carried out through acid precipitation in polyol medium which functioned as a stabilizing agent.[66]

Table 1 Some important synthetic methods for lignin NPs and their related details

Lignin	Size (nm)	Materials and conditions	Advantages or disadvantages	Ref.
Steam exploded rice straw	15–20	Castor oil, ethylene glycol and water; acid precipitation in polyol medium which functioned as a stabilizing agent.	Covered 9 out of 12 green chemistry principles; environmentally-friendly and facile approach.	66
Alkaline; pretreated; enzymatic hydrolysis; Kraft	80–230	Self-assembly in recyclable and non-toxic aqueous sodium *p*-toluenesulfonate at RT, with a lowest concentration of up to 48 g L^{-1}	The unfavorable factors *e.g.* processing pH and lignin species were eliminated using hydrotropic chemistry and the synergistic dissociation of the entrapped *p*-toluenesulfonate and intrinsic phenolic hydroxyl and carboxylic acid moieties of the lignin NPs; high yields	73
Alkali	≥100	Lignin *via* a microwave-(MW) assisted acetylation process using acetic anhydride under catalyst- and solvent-free conditions; acetic anhydride serving as both a reagent and dispersion solvent.	The NPs were rapidly produced without dialysis and separated by centrifugation; the applied solvent could be recycled and reused, which reduced cost amenable for industrial scale-up production; facile and relatively green method.	74
Kraft	~38	One-pot *via* a compressed CO_2 antisolvent; *N*,*N*-dimethylformamide to produce lignin solution.	High Brunauer–Emmett–Teller surface area (~92 m^2 g^{-1}) comprising mesopores with relatively higher UV absorbing and dispersion stability, accelerated solubility, and homogeneous thermal degradation activity; facile and environmentally friendly method.	68
Waste obtained from Kraft pulping	200–500	Water as a non-solvent reducing lignin's degrees of freedom causing the segregation of hydrophobic regions to compartments within the produced NPs; NPs were prepared during the dialysis process, 24 hours under slow stirring in a fume hood.	Simple process with no chemical modification needed; the NPs were found to be surface modifiable *via* adsorption of oppositely charged poly-electrolytes, potentially for scale-up.	67
Natural lignocellulose source of rice straw	~100	Ethanol–water co-solvent system with an acid catalyst by a one-step MW-assisted method.	Synchronous and rapid preparation (10 min); highly monodisperse and stable lignin NPs with functional groups with great potential for biomedical applications.	72
Kraft and organosolv	45–250	Flash precipitation of the dissolved lignin polymer; the colloidal synthesis and interfacial design of NPs with tunable surface characteristics	Fine-tunable surface characteristics *via* coating with a cationic polyelectrolyte; control of the surface charge of NPs and their stability in strongly basic regimes with applications under extreme pH conditions; formulation of an NP platform for the next generation of sustainable delivery nanovehicles for antimicrobials, fungicides, pesticides, and drugs. Up-scalable and sustainable method.	75
Wheat straw (herbaceous biomass)	70–90	Choline chloride-lactic acid-based deep eutectic solvent extraction of lignin NPs from herbaceous biomass.	Highly pure and well-dispersed lignin NPs were prepared in high yields.	76
Bio-refined comprising grafted carbohydrates	~100	NPs from a carbohydrate-rich fraction of lignin (24% by weight) as the insoluble fraction in 0.5 M NaOH. Dissolution in an NaOH/urea/water system with dropwise addition to water under a high shear, which rapidly produced precipitated particles.	Green and sustainable method; highly stable particles. The NPs, containing both carbohydrates and lignin, can be applied for exclusive applications due to their bi-component nature.	77

Lignin NPs have been synthesized, with a hydrodynamic diameter ranging from *ca.* 80 to 230 nm, *via* self-assembly in a recyclable and non-toxic aqueous sodium *p*-toluenesulfonate (*p*-TsONa) solution at room temperature, with a lowest concentration of up to 48 g L^{-1}. The drug-encapsulated lignin NPs demonstrated improved characteristics, with sustained drug-releasing capability and biocompatibility. It appears that the characteristics of the lignin NPs depend on the synergistic dissociation of the entrapped sodium *p*-toluenesulfonate and the intrinsic phenolic OH and COOH groups of the lignin NPs. Additionally, the unloaded drug and free sodium *p*-toluenesulfonate could be recycled for multiple applications, thus contributing to environmental sustainability (Fig. 2 and 3).[73]

Lignin has been modified *via* a microwave (MW)-assisted acetylation procedure by applying acetic anhydride under catalyst- and solvent-free conditions, with acetic anhydride serving as both a reagent and a dispersion solvent. Subsequently, the regular lignin NPs were produced in high yield by solvent shifting combined with an ultrasound (US) process. The lignin NPs were rapidly generated without dialysis and easily separated by centrifugation; meanwhile, the applied solvent could be recycled and reused, which would simplify the process, limit the cost, and realize industrial scale-up production. The size of the lignin NPs can be varied with the preparation conditions, including the lignin characteristics, the initial concentration of lignin, and the ultrasonic intensity. The highest yield of lignin NPs reached was about 82% as the initial concentration

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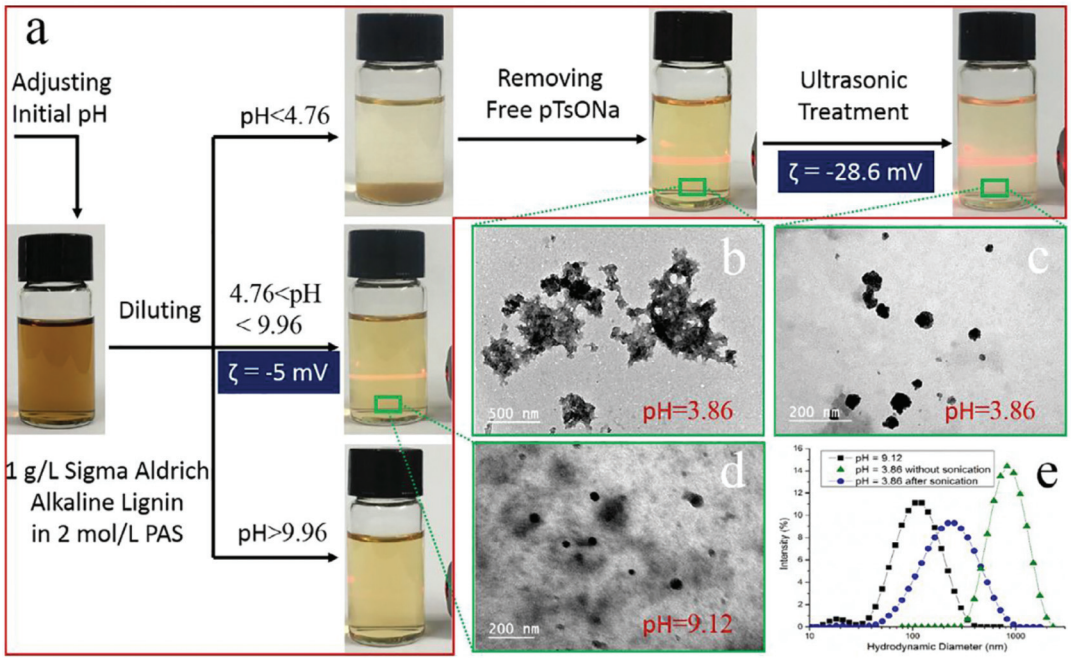


Fig. 2 (a) Schematic illustration of lignin NP production, (b–d) transmission electron microscopy (TEM) images of lignin NPs from solutions of various initial pH values, and (e) size distribution of lignin NPs. Reproduced with permission from ref. 73.

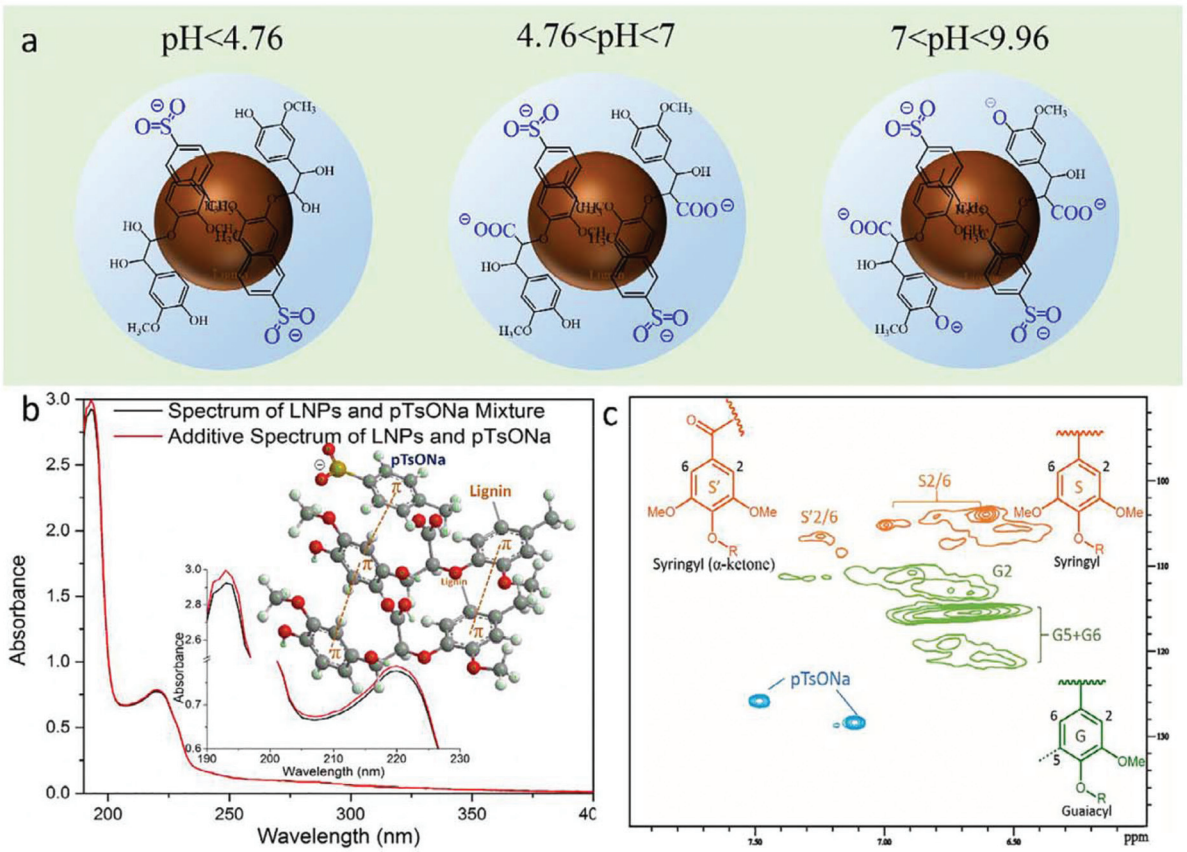


Fig. 3 (a) Schematic illustration of lignin NP stabilities in solutions of various initial processing pH values, (b) UV-Vis additive spectrum of lignin NPs and *p*-TsONa and the spectrum of the lignin NP and *p*-TsONa mixture, and (c) 2D heteronuclear single quantum coherence (HSQC) NMR spectrum of the resultant lignin NPs from Kraft lignin. Reproduced with permission from ref. 73.

of lignin and the US intensity were increased; US treatment improved the uniformity and dispersion of the NPs.[74]

In one study, citric acid modification of two different lignocellulosic products, cellulose nanocrystals (CNC) and lignin NPs, was accomplished. The resultant modified cellulose nanocrystals exhibited improved dispersion in polar solvents and better thermal stability as compared with cellulose nanocrystals while, in the case of lignin NPs, a slight increase in thermal stability and alteration of the dispersibility of modified lignin NPs in polar solvents were discerned. These results affirmed that bio-based sustainable NPs with tuned hydrophilicity namely esterified (amended cellulose nanocrystals) and etherified (modified lignin NPs) were obtainable at a low cost, with a readily dissolvable cross-linker, and have potential for widespread applicability in the field of polymer-based nanocomposites encompassing varying polarity.[78] Additionally, lignin NPs have been esterified and etherified by citric acid and then blended with poly(vinyl alcohol) (PVA) at different loading levels (up to 10 wt%); PVA nanocomposite films containing unmodified and modified lignin NPs have been produced *via* the solvent casting method.[79] Microscopic evaluations of fractured surfaces of films demonstrated that modified lignin NPs could be evenly dispersed in the PVA matrix; no macroscopic phase separation ensued. The water contact angle, swelling efficiency and moisture adsorption properties of unmodified and modified lignin NP based films were evaluated and compared, demonstrating an overall acceleration of dimensional stability and decreased moisture adsorption for modified lignin NP nanocomposites; they outperformed the films containing only native lignin NPs in terms of the antioxidant property and transparency. With slight compromise in the UV-absorbing property, these materials can be promising candidates for applications necessitating high antioxidant potential and enhanced performance towards light absorbance.[79]

NPs from commercial Kraft lignin were developed using facile and one-pot green technology using a compressed CO_2 anti-solvent; *N*,*N*-dimethylformamide (DMF) was employed as an organic solvent to produce the lignin solution. It was shown that as the temperature and lignin concentration increased, the pressure and solution flow rate reduced. Furthermore, the degree of particle aggregation/coalescence and the size increased along with the broader size distribution. Particularly, the coalescence of particles was meaningfully influenced by the operational pressure, and even more substantially with the increasing temperature. Consequently, uniform, quasi-spherical NPs (about 38 nm) were produced at 280.2 K, 15.0 MPa, and 0.06 kg h^{-1} of the solution flow rate and 5.3 wt% of the initial lignin concentration. Additionally, the lignin NPs showed a relatively high Brunauer–Emmett–Teller (BET) surface area (~92 m^2 g^{-1}) which mainly consisted of mesopores, and demonstrated higher UV absorbing and dispersion stability, accelerated solubility, and homogeneous thermal degradation activity when compared to the raw lignin.[68]

An approach to produce lignin NPs from waste lignin obtained from Kraft pulping has been reported, wherein spherical lignin NPs were obtained by dissolving softwood Kraft lignin in tetrahydrofuran and subsequently introducing water into the system through dialysis; no chemical modification of lignin was required. Water functioned as a non-solvent reducing lignin's degrees of freedom causing the segregation of hydrophobic regions to compartments within thus forming NPs. The final size of the NPs depended on the pre-dialysis concentration of dissolved lignin. The stability of the nanoparticle dispersion as a function of time, salt concentration and pH has been studied. In pure water and at room temperature, the lignin nanoparticle dispersion was stable for over two months, but a very low pH or high salt concentration induced aggregation. It was further revealed that the surface charge of the particles could be reversed and stable cationic lignin NPs were prepared by the adsorption of poly(diallyldimethylammonium chloride).[67]

The synchronous and rapid production of spherical lignin NPs (~100 nm) along with carbon quantum dots (~2–3 nm) using a one-pot MW-assisted strategy has been reported.[72] The produced lignin NPs having a uniform roughly spherical shape, with monodisperse characteristics, showed good stability.[72] The dissolved lignin fraction under thermochemical treatment formed lignin droplets through a micellization procedure with water. Next, the reduction in solvent polarity facilitated the dispersion of lignin droplets in solution and the fabrication of lignin NPs ensued *via* a layer-by-layer self-assembly method. Additionally, the synchronously produced nitrogen-doped carbon quantum dots with prominent up-conversion photoluminescence demonstrated significant potential for bioimaging.[72]

The colloidal synthesis and interfacial design of NPs with tunable surface characteristics have been reported by using two lignin precursors, Kraft and organosolv lignin.[75] The green synthesis method was based on flash precipitation of a dissolved lignin polymer, which enabled the production of NPs (~45–250 nm). The size evolution of the two types of lignin particles was fitted on the basis of modified diffusive growth kinetics and mass balance dependency. The surface characteristics of the NPs were fine-tuned by coating them with a cationic polyelectrolyte, poly(diallyldimethylammonium chloride). Additionally, it was revealed that the properties of NPs were influenced and regulated by the type of applied lignin and the existence of a polyelectrolyte surface coating which allowed the control of the surface charge of NPs and the extension of their stability to significant basic regimes, thus enabling their potential applications under extreme pH conditions.[75] Furthermore, highly pure and well-dispersed lignin NPs (~70–90 nm) were isolated from wheat straw (herbaceous biomass) with yields up to 81.5% from air-dried samples and 85.9% from oven-dried samples by choline chloride-lactic acid-based deep eutectic solvent (DES) extraction.[76] It was revealed that the extraction temperature, time, and water content in wheat straw showed meaningful impact on the lignin characteristics, including monomeric types, ether bond linkages, and NP size. A mechanistic understanding of the DES deconstruction of wheat straw and synthesis of purified lignin NPs has

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been documented. Accordingly, DES can cleave both the ether bonds in lignin and linkages between lignin and the hemicellulose.[76]

Wood waste (*e.g.*, sawdust) can be applied for developing newer wood waste-based products, instead of its exclusive application in energy generation. The wood industries can lower transport costs and decrease the environmental impact of their overall fabrication procedure *via* the implementation of new value-chains.[80] Acidolysis lignin fractions have been isolated from sawdust of Iroko and from mixed sawdust of Iroko and Norway spruce, and they have been structurally characterized prior to their application in producing lignin NPs by a non-solvent method.[81] Consequently, aqueous dispersions of lignin NPs have been applied for surface treatment of beech wood *via* dip-coating and their stability to artificial weathering has been evaluated. For this study, the coated samples and homologous non-coated samples underwent an artificial weathering treatment in a UV chamber which is the most common test of resistance to weathering degradation. Lignin NPs or colloidal lignin particles have already been used for tuning commercial wood stains.[80]

Though, for constructing an economically feasible integrated lignocellulose biorefinery, the conversion of a low value but high-volume lignin by-product to its usable form is one of the determinant parameters, its structural complexity and inhomogeneity are truly challenging.[82] In one study, uniform lignin NPs have been produced from various technical lignins, despite the varied lignocellulosic biomass and the pretreatment methods/conditions applied. It appears that, value-added nanostructure lignin enriched with multifunctional groups can be a promising versatile material platform for various downstream uses, especially in the emerging nanocomposite fields. Two types of uniform lignin NPs have been produced *via* self-assembling of a deep eutectic solvent and ethanol-organosolv extracted technical lignins derived from a two-stage fractionation pretreatment method, respectively (Fig. 4). Both lignin NPs showed a spherical morphology with an exclusive core–shell nanostructure, but the deep eutectic solvent-lignin NPs exhibited a more uniform particle size distribution. When incorporated into the traditional polymeric matrix such as PVA, these lignin NP products showed significant potential to formulate a transparent nanocomposite film with additional UV-shielding efficacy (reaching ~80% at 400 nm with 4 wt% of lignin NPs) and antioxidant functionalities (attainment of ~160 μm mol Trolox per g with 4 wt% of lignin NPs). Additionally, the abundant phenolic hydroxyl

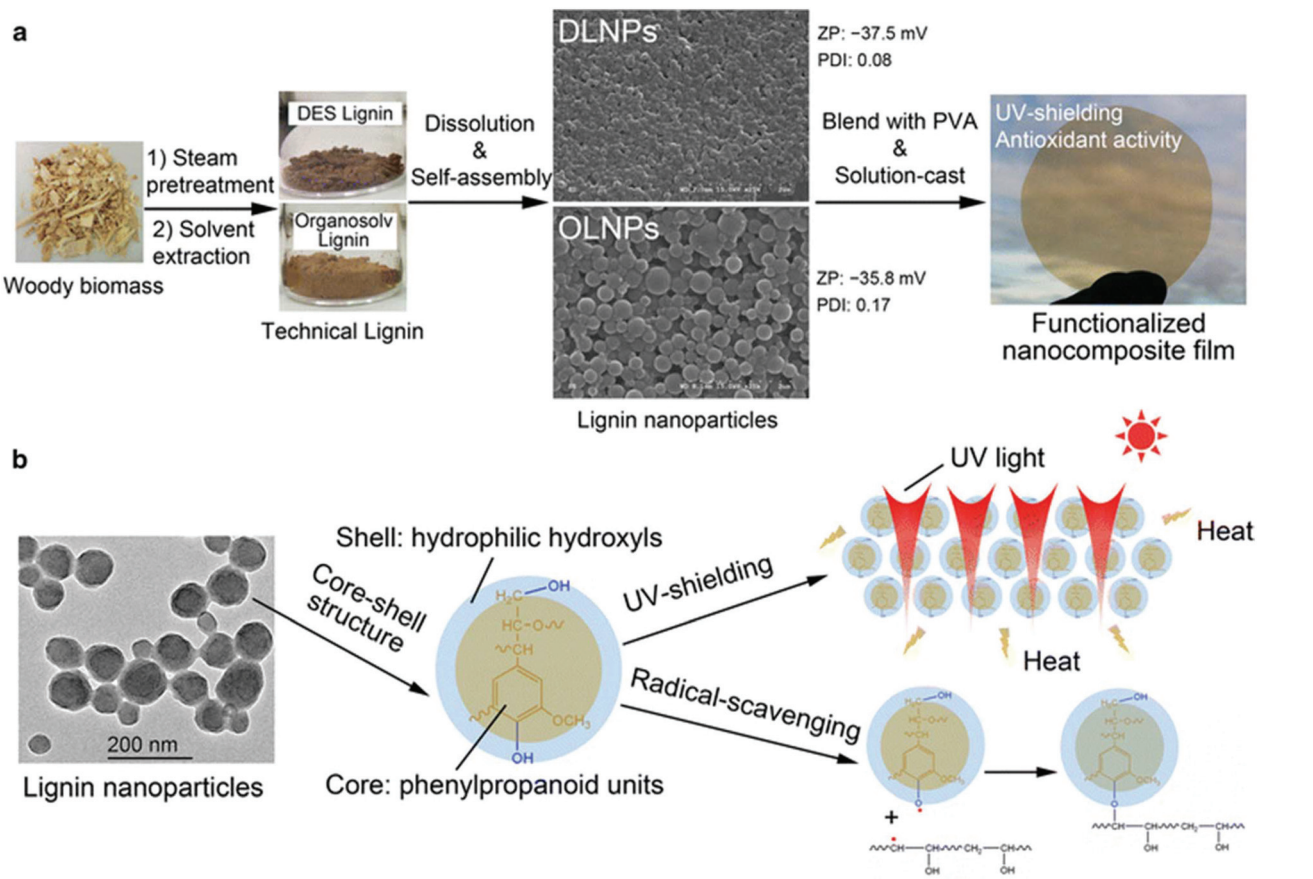


Fig. 4 (a) Production of lignin NPs and a lignin NP/PVA composite film. ZP: zeta-potential value, PDI: polydispersity index. (b) Suggested mechanism of UV-shielding and antioxidant activity by using lignin NPs as the functional additive. Reused from ref. 82, An Open Access Article (CC BY).

groups on the shell of lignin NPs offered acceptable interfacial adhesion with the PVA matrix *via* the production of a hydrogen bonding network, which additionally enriched the mechanical and thermal performances of the produced nanocomposite films.[82]

Solvent exchange and pH modifications can stimulate the self-assembly of lignin bio-polymers, and permit the generation of suitable functional nanomaterials.[83] Indeed, the rational design of these materials needs an in-depth consideration of the underlying lignin self-assembly in solution. Colloidal transformations from nano-sized lignin assemblies to submicron-sized spherical particles upon solvent exchange have been investigated. The surface fractal structure and stability of these particles were found to be significantly dependent on the solvent and pH, with aggregation to a gel-like material at low pH. Analysis confirmed colloidal transformations from about 6 nm polymer assemblies of an approximately ellipsoidal shape to well-defined sub-micron particles upon solvent exchange.[83] These nanostructured lignin-based functional materials can be applied in various fields of biomedicine and food science. In another study, a production technique of lignin-based microstructures or nanostructures through self-assembly was exploited.[84] Consequently, assorted self-assembled nanostructures have been prepared, including nanocubes, nanorods, nano-octahedra, nanobranches, and other morphologies, by selecting various types of simple inorganic acids at different pHs. The produced structures have been fully analyzed; they were uniform and stable, with little or no cytotoxicity to cells *in vitro* at concentrations up to 1024 μg mL^{-1}, reaffirming that these NPs are cell biocompatible and biodegradable. This study provided insights into the mechanism of self-assembly of lignosulfonate NPs and paved the way for their further development in the fields of bio- and nano-medicine.[84]

4. Lignin: immense valuable applications

Lignin can be applied for various industrial and biomedical applications, including chemicals, biofuels, and polymers, and the development of nanomaterials for drug delivery but these applications depend on the source, chemical modifications and physicochemical properties.[85,86] Lignins can be used in medicine and pharmaceutics to improve human health because of its inherent antioxidant and antimicrobial characteristics. Additionally, lignins show biological activities namely the capability of reducing cholesterol by binding to bile acids in the intestine. Such activities of lignins open windows of opportunities in the treatment of various diseases such as obesity, diabetes, thrombosis, viral infections and cancers (Fig. 5).[14] The structural and functional modifications of lignin can increase its antidiabetic and antioxidant activities. For instance, alkali lignin, extracted from the deciduous plant *Acacia nilotica*, showed antioxidant and antidiabetic characteristics;[87] this modified alkali lignin exhibits α-amylase inhibitory activities and anti-hyperglycemic characteristics, which makes it as an appropriate candidate for healing diabetes.[87] Lignosulfonic acid (a derivative of lignin) is a non-competitive inhibitor of α-glucosidase, and has a capability of causing a delay in glucose absorption. This suggests that lignin and its derivatives, mainly lignosulfonic acid, can be used for the treatment of carbohydrate absorption and related diseases, such as diabetes.[88]

Kraft lignin is identified as a fat adsorbent which can control and prevent obesity. Lignophenols, as stable and antioxidant derivatives of lignin, have the capability of reducing cholesterol levels in HepG2 cells in a dose-dependent manner.[89] An experiment with rats which were fed a high dose

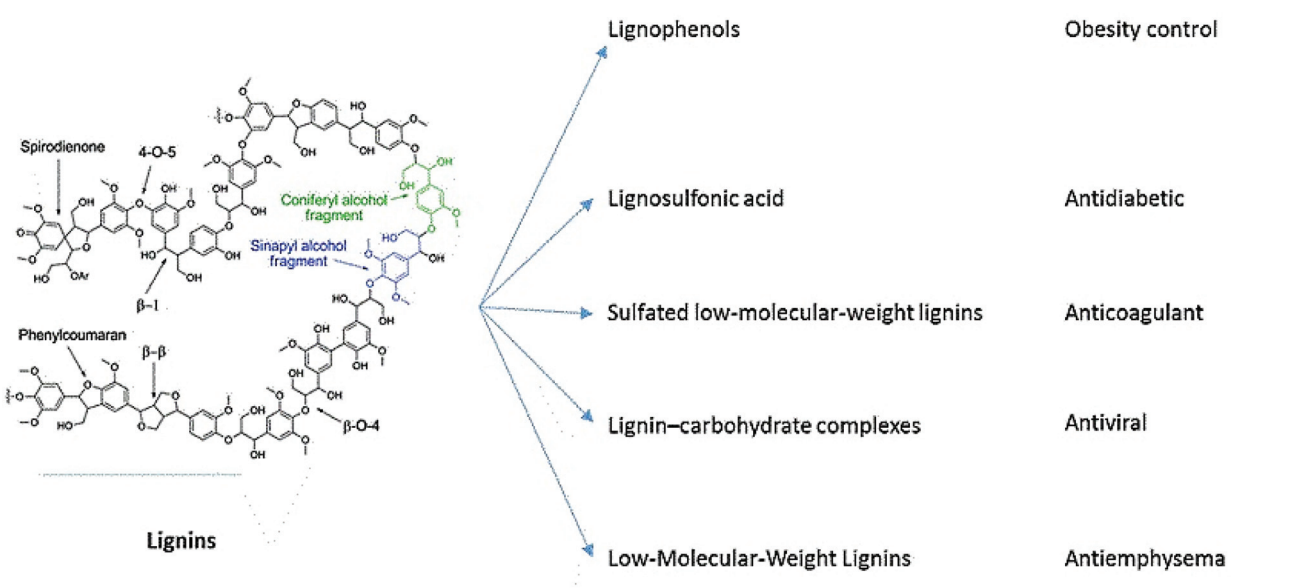


Fig. 5 Beneficial effects of lignins and their derivatives. Reused from ref. 14, An Open Access Article (CC BY).

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fat diet, has shown that this lignin could suppress adipose tissues, reduce the triglyceride level in plasma, and attune the hepatic expression of SREBP-1c mRNA. However, the exact mechanism of LP effects on lipid metabolism and obesity control still needs to be further addressed.[90] Additionally, investigations have uncovered the antiviral activity of lignin–carbohydrate–protein complexes (LC) against some viruses such as herpes simplex virus types 1 and 2 (HSV-1 and -2), human cytomegalovirus (HCMV) and measles virus.[91] Lignosulfonic acid, which is mostly used in the paper industry and is involved in the formation of an artificial vanilla flavor,[92] shows anti-HIV activity when used in low dosage.[93] Its potential therapeutic role as an anti-viral agent has been approved in clinical trials with established safety in the prevention of sexual transmission of HIV-1.[94] Furthermore, sulfated low-molecular-weight lignins (LMWLs) have been shown to possess anticoagulant and immunomodulatory mediator properties; they comprise oligomeric chains of different lengths and different inter-monomeric linkages, such as β-*O*-4 and β-5.[95] Their interaction with the heparin binding domain inhibits a variety of serine coagulation proteases. Although some other ligands have the ability to bind to the heparin binding site of serine coagulation proteases, none has direct anti-coagulant effects.[96] Sulfated β-*O*-lignins also have anti-coagulant properties and hence they can concurrently stimulate anti-coagulation and anti-platelet effects.[97]

4.1. Lignin for eco-friendly synthesis of metallic NPs

Lignin can be used for greener synthesis of metallic NPs. For instance, silver NPs were produced from aqueous silver nitrate with alkali lignin (low sulfonate) playing a dual role, as a reducing and capping agent; the synthesis mechanisms were highly pH dependent. Under neutral and acidic conditions, poly-disperse NPs were produced *via* the self-catalyzed reduction of Ag^+ on instantaneously formed Ag_2O surfaces followed by the slower pseudo-first order reduction. The Ag_2O NPs functioned as the nucleating sites for the reduction of the remaining silver cations to form silver NPs whose size and size distribution were strongly dependent on the lignin concentration. The NPs were optimally synthesized by reducing 2 mmol L^{-1} silver nitrate with 0.16 wt% alkali lignin (low sulfonate) at pH 10 and 85 °C in 30 min to afford nearly quantitative yield in bimodal distributed sizes with 23% and 77% in Feret diameters of 7.3 (±2.2) nm and 14.3 (±1.8) nm, respectively.[98] In another study, lignin was used for producing silver NPs, which were incorporated into agar-based films; such amalgamation of lignin and silver NPs accelerated the mechanical, UV-light barrier, and water vapor barrier properties of the composite films. The loading of 1 wt% of silver NPs was the optimum concentration to improve the mechanical and water vapor barrier properties of the films; the produced films containing silver NPs showed antibacterial activity against food-borne pathogenic bacteria, *Escherichia coli* and *Listeria monocytogenes*.[99] Additionally, in a similar study, organosolv lignin was used as a reducing agent for the production of silver NPs and their incorporation into a poly(lactide) (PLA) polymer to generate composite films; such silver NP-bearing films displayed potent antibacterial activity against *E. coli* and *L. monocytogenes*.[100]

Water insoluble alkali lignin has been deployed in a ball milling method for crushing lignin, wherein its specific surface area increased from 1.9 to 10.9 $m^2\ g^{-1}$; the alkali lignin after milling was used for reducing metal ions. It was illustrated that alkali lignin and hemicellulose could be used as efficient reducing agents to produce Pt, Au and Pd NPs, and at the same time, the alkali lignin could be considered as a useful stabilizing agent to disperse the acquired metal NPs.[101] In another study, broccoli-shaped Cu_2O NPs were produced by an eco-friendly, gentle and low-cost synthetic method using lignin as a reducing and capping reagent; the morphology of the ensuing Cu_2O NPs appeared as a loose accumulation of particles just like broccoli, with the particle size range being 100 and 200 nm. The produced NPs demonstrated effective bactericidal activity against *E. coli* and *S. aureus*; the antibacterial rate could be 100% after 30 min with 4.0 g L^{-1} Cu_2O NPs. Interestingly, the produced NPs had low cytotoxicity.[102]

4.2. Biomedical applications of lignin NPs

4.2.1. Drug delivery. Numerous reports have shown the capacity of lignin-based NPs for controlled drug release which is important in medicine.[103] NPs synthesized from lignin are relatively non-toxic, highly biodegradable, stable and inexpensive; these four major advantages embody them as potent drug delivery systems in human diseases (Table 2).[67,104] Recently, water-dispersed lignin NPs have been made to carry silver ions, which are significant in antimicrobial applications or in cancer treatments.[104] Additionally, lignins are appropriate for transferring both hydrophobic and hydrophilic drugs. For instance, the pH-sensitive polymers are added to lignin NPs for facilitating easier loading of hydrophilic drugs.[75] Poorly water soluble drugs or water-soluble anticancer drugs can be loaded on lignin NPs, leading to an increase in their anticancer and growth inhibitory effects in different cancer cell lines.[105]

Lignin has also been applied for transferring an arthritis rheumatoid specific drug, methrotrexate, in a rat model wherein it could release the drug into blood vessels and inflamed tissues.[111] The capacity of lignin, in drug delivery systems, is being currently developed to introduce it as a suitable candidate for further usage in medicine.[112] In one study, pure lignin NPs, iron(III)-complexed lignin NPs, and Fe_3O_4-infused lignin NPs have been produced with a round shape, narrow size distribution, reduced polydispersity and good stability at pH 7.4. It has been reported that the produced lignin NPs had low cytotoxicity in all the tested cell lines and hemolytic rates below 12% after incubation of 12 h. Pure lignin NPs demonstrated the capacity to robustly load poorly water-soluble drugs and other cytotoxic agents, including sorafenib and benzazulene, and improved their release profiles at pH 5.5 and 7.4 in a sustained manner. Additionally, benzazulene-pure lignin NPs exhibited an accelerated anti-proliferation influence in various cells compared to the pure benzazulene,

Table 2 Lignin NPs: some important examples of drug delivery applications

Matrix type	Encapsulation approach and active ingredients	Findings	Ref.
Lignin NPs from indulin AT	NP flash precipitation with subsequent silver ion infusion and polyelectrolyte coating	>95% release of silver ions in 24 hours and antibacterial influence against *E. coli*, *P. aeruginosa* and *Rastonia* sp.	106
Lignin NPs from alkaline lignin	Incorporation of Resveratrol® during particle formation through a polarity change	Approximately 80% drug released into phosphate buffer saline (PBS) after 4 days	107
Lignin NPs from LignoBoost™ softwood Kraft lignin	Incorporation of poorly water-soluble Sorafenib® and Benzazulene® during particle formation *via* a polarity change	Poorly water-soluble drugs are released upon degradation of the particles; the water-soluble drug could not be incorporated into the nanoparticle; low cytotoxic effects on cancer cell lines: MDA-MB-231, MCF-7, PC3-MM2, Caco-2 and non-tumor cells: KG1 and EA·hy926 endothelial cells	83
Polyelectrolyte microparticles of quaternary ammonium lignin–sodium dodecyl benzenesulfonate (lignin from pine alkali lignin)	Loading of hydrophobic Avermectine during particle precipitation	Release of ~80% Avermectine into methanol : water (1 : 1) after 72 h; good UV protection of the drug (85% preserved after 96 h UV irradiation 30 W, 310 nm)	108
Crosslinked cellulose-lignin hydrogels (steam expansion lignin, aspen wood)	Swelling of the gel in polyphenol solution	A higher lignin content leads to a faster drug release, up to 30% in 10 hours	109
Lignin droplets in W/O Pickering emulsion coated with polyurea	Loading of hydrophobic Avermectine in the emulsion before droplet coating reaction	Release of 85% of Avermectine into 4 : 1 ethanol : water after 72 hours; lignin-polyurea coatings were more porous than pure polyuria layers, which showed a more sustained release; UV protection of lignin coatings was good (>75% preserved after 120 h irradiation, 30 W, 310 nm)	110

and displayed maximum inhibitory concentrations ranging from 0.64 to 12.4 μM after 24 hours of incubation (Fig. 6).[105]

The alkali lignin was used for producing NPs with perfect spheres and good dispersibility through a simple self-assembly approach by adding water to a methanolic solution of alkali lignin. Self-assembly of alkali lignin with the bioactive molecule resveratrol and Fe_3O_4 NPs led to the preparation of a stable nano-drug carrier. From cytological and animal analyses, it was found that the magnetic resveratrol-loaded lignin NPs showed significant anticancer effects, and accelerated *in vitro* resveratrol release and stability, drug accumulation, and suitable tumor reduction, and showed lower adverse effects than free drugs (Fig. 7).[107] In another study, a lignin-based targeted polymeric NP platform, folic acid–polyethylene glycol–alkaline lignin conjugates, was produced through self-assembly for delivery of an anticancer drug (hydroxyl camptothecin, HCPT). The lignin-based NPs (about 150 nm) showed outstanding biocompatibility, high drug loading efficiency, prolonged blood circulation time, and improved cellular uptake; these generated systems can be used as a promising candidate for anticancer drug delivery (Fig. 8).[113]

The preparation of targeted lignin-based drug delivery NPs has been accomplished for loading doxorubicin hydrochloride. The lignin hollow NPs have been applied as a platform for the production of targeted delivery materials by incorporating magnetic NPs and folic acid through layer-by-layer self-assembly. It has been reported that the surface of lignin hollow NPs was covered uniformly by Fe_3O_4 NPs and grafted with folic acid. The folic-magnetic-functionalized lignin hollow NPs could respond to a magnetic field and folic acid receptors. Additionally, the targeting performance of the folic-magnetic-functionalized lignin hollow NPs accelerated the cellular uptake of NPs in the case of HeLa cells (Fig. 9).

Lignin NPs have been prepared as an oral drug delivery system for curcumin. The particle size of curcumin-loaded lignin NPs was about 104 nm, and the encapsulation efficiency of curcumin in the NPs was 92%. It was revealed that curcumin-loaded lignin NPs had appropriate stability in a simulated gastric fluid and showed slow release under intestinal conditions as desirable. *In vivo* pharmacokinetic evaluation showed that the lignin NP system accelerated the bioavailability of curcumin by ten-fold compared to the administration of unformulated curcumin.[115] Additionally, carboxylated lignin NPs have been functionalized with a block copolymer made from PEG, poly(histidine) and a cell-penetrating peptide for targeted and pH-responsive delivery of anticancer drugs. The prepared NPs had spherical shapes and improved size distributions, good stability in physiological media and low cytotoxicity in all the tested cell lines. A poorly water-soluble cytotoxic agent was successfully loaded into the carboxylated lignin NPs, refining its release profiles in a pH-sensitive manner and exhibiting an accelerated anti-proliferative influence in various cancer cells compared with a normal endothelial cell line.[116] In another study, the slightly water-soluble gatifloxacin, readily water-soluble doxorubicin hydrochloride, and water-insoluble doxorubicin were selected for investigation of their loading capability and delivery profile of the produced lignin NPs. Consequently, it was shown that the lignin NPs, prepared by using aqueous sodium *p*-toluenesulfonate solution, had significant loading capability for gatifloxacin, doxorubicin hydrochloride and doxorubicin. The biocompatibility of the lignin NPs and the antitumor effect of the

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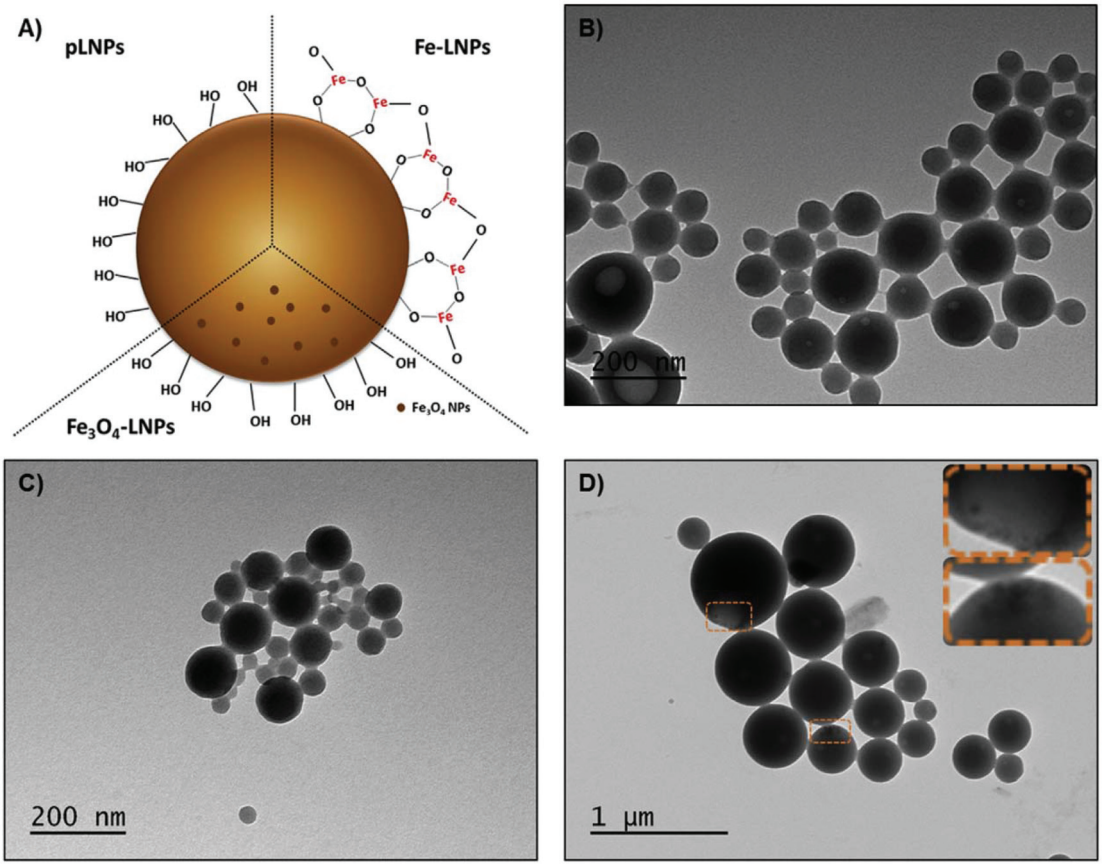


Fig. 6 Three kinds of lignin NPs: schematic representation (A), TEM images of (B) pure lignin NPs, (C) iron(III)-complexed lignin NPs and (D) Fe_3O_4-infused lignin NPs and magnification of the Fe_3O_4 NPs inside the lignin NPs. Reproduced with permission from ref. 105.

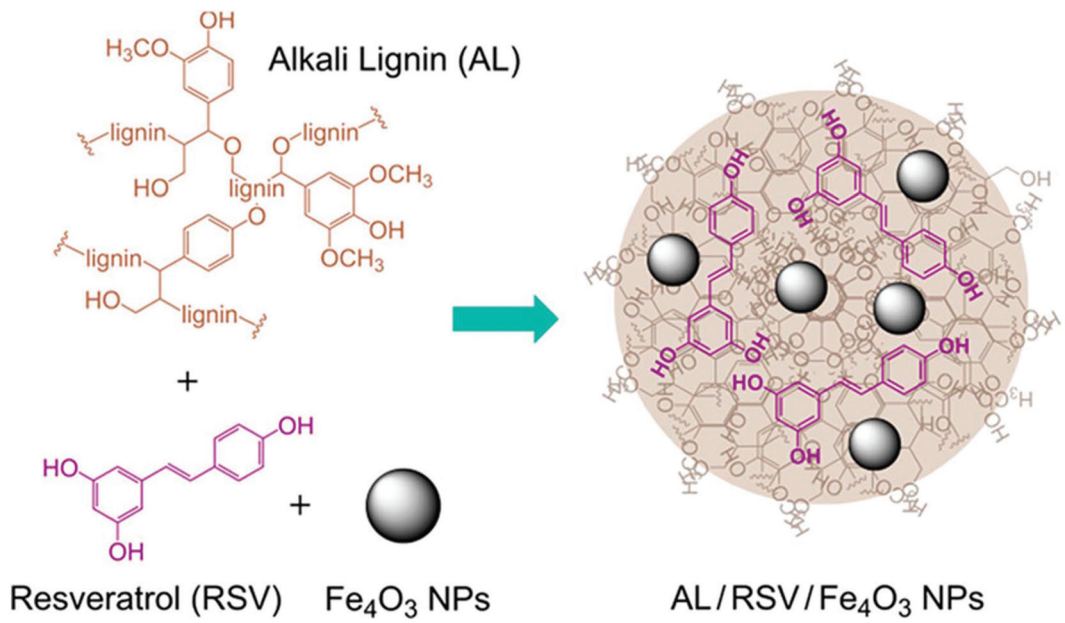


Fig. 7 Representation of the self-assembly procedure for alkali lignin/resveratrol/Fe_3O_4 NPs. Reproduced with permission from ref. 107. Copyright (2017) American Chemical Society.

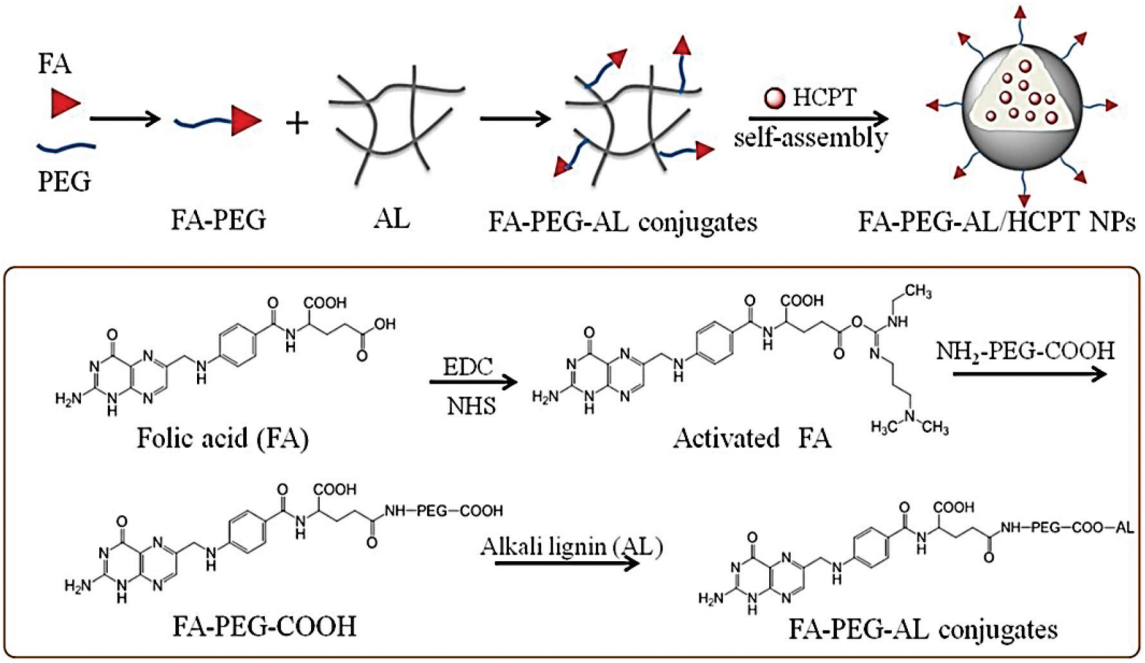


Fig. 8 Schematic illustration of the self-assembly procedure for folic acid–polyethylene glycol–alkaline lignin conjugates (FA-PEG-AL)/hydroxyl camptothecin (HCPT) NPs and synthetic procedure for FA-PEG-AL conjugates. Reproduced with permission from ref. 113. Copyright (2018) American Chemical Society.

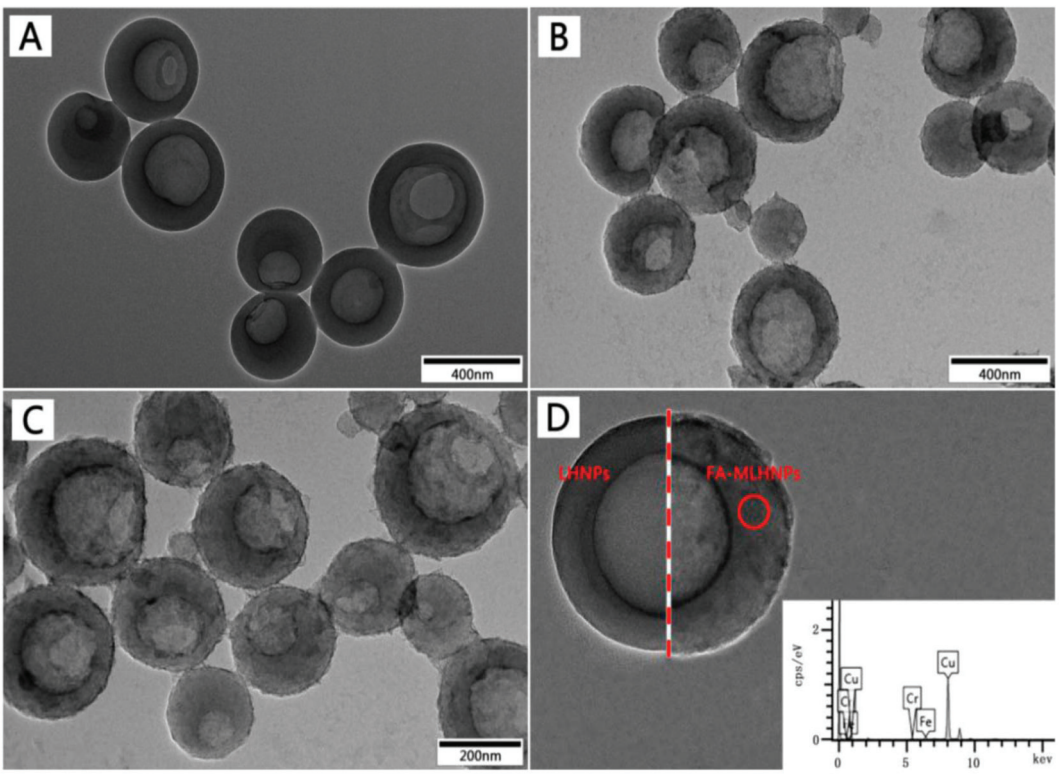


Fig. 9 TEM images of (A) lignin hollow NPs, (B) magnetic-functionalized lignin hollow NPs, (C) folic-magnetic-functionalized lignin hollow NPs, and (D) lignin hollow NPs compared, inset is the energy spectrum of the surface for folic-magnetic-functionalized lignin hollow NPs. Reused from ref. 114, An Open Access Article (CC BY).

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doxorubicin-encapsulated lignin NPs were preliminarily analyzed *in vivo*. No substantial histologic differences between the lignin NP and phosphate-buffered saline control groups were detected, further indicating the good biocompatibility of the resultant lignin NPs, which demonstrated no side effects unlike free doxorubicin.[73]

Water-soluble reduced fluorescent carbon dots (r-FCDs) were produced by using lignosulfonate lignin as a carbon source and they were deployed as nano-carriers for curcumin and bio-imaging applications; r-FCDs were obtained within 10 min under MW irradiation after reduction with $NaBH_4$. The weak interaction found between an anticancer compound curcumin and surfaces of r-FCDs endowed them with the capacity to act as a superior candidate for its capture and release at physiological pH (7.4). The r-FCDs were efficiently taken up by cancerous cells (A549 and SW480) illuminating the whole cell with a clear distinction between the cytoplasm and nucleus. Good biocompatibility results have been reported with MTT and hemolysis assays; the prepared carbon dots can be used as a promising probe for cancer diagnosis and treatment.[117] In a similar study, water-soluble and eco-friendly fluorescent carbon dots were prepared using the renewable resource lignin as a precursor, a waste emanating from the pulp and paper industries and lignocellulosic biofuel production. The protocol used a compressed liquid CO_2 anti-solvent method followed by carbonization and chemical oxidation with an acid mixture ($H_2SO_4 : HNO_3$ = 1 : 3, v/v) at 90 °C for 8, 16, and 24 h. The carbon dots showed excellent stability in a highly concentrated NaCl solution, photostability with pH-dependent FL emission activities, and cellular imaging capability with very low cytotoxicity. Notably, the FL emission intensities of the as-synthesized carbon dots increased with longer reaction times, indicating a high degree of oxidation on the carbon dot surface and, consequently, a high quantum yield of 13% ensued. Thus, it appears that the produced carbon dots can be used as promising fluorescent probes for bioimaging applications *in vitro* and *in vivo*.[118]

Oil-filled microcapsules of Kraft lignin were produced by first creating an oil in water emulsion followed by a high-intensity, ultrasound-assisted cross-linking of lignin at the water/oil interface. The ability of lignin microcapsules to incorporate and release Coumarin-6 was analyzed. The ensuing lignin microcapsules were not cytotoxic and readily incorporated into Chinese hamster ovary cells.[119] In another study, the concerted use of chelation chemistry, oil in water emulsion principles, and low energy sonication, offers a facile, one-pot strategy to assemble lignin nanocapsules of a controlled architecture. Under these conditions, capsules were shown to rapidly assemble utilizing two driving forces, the π-stacking propensity of lignin and its metal chelating ability at alkaline pH. Consequently, pure lignin nanocapsules and Fe–lignin nanocapsules were produced in high yields with size distributions varying from 0.3 to 6 μm and their release efficiencies were analyzed. It has been reported that the complexation effects of the phenolic OH groups offered the Fe–lignin nanocapsules increased stability, reduced shell thickness (allowing for greater loading efficiencies), and lowered release kinetics, compared to lignin nanocapsules. In yet another study, cross-linked and decolorized lignin NPs have been produced enzymatically and chemically from softwood Kraft lignin. The research findings have high impact in the utilization of Kraft lignin as nano-sized colloidal particles in advanced bio-nanomaterial applications in medicine, foods and cosmetics including diverse chemical industry sectors.[120]

Eco-friendly biodegradable lignin NPs were produced using two different methods which were found to be non-toxic for yeast and microalgae.[104] The first approach involved the precipitation of low-sulfonated lignin from an ethylene glycol solution deploying various acidic aqueous dilute solutions; the obtained NPs were found to be stable over a wide pH range. The second approach used high-pH aqueous solution for the acidic precipitation of lignin, wherein the generated NPs were found to be stable only at low pH. Such particles have potential for applications as drug delivery vehicles and stabilisers in the cosmetic and pharmaceutical industries.[104] In order to understand the influence of the solution structure, especially hydrogen bonding of ethylene glycol on the synthesis of lignin NPs, dynamic light scattering and small angle neutron scattering studies were accomplished.[121] Nano-sized lignin colloidal spheres have been synthesized using a solution of acetylated lignin with a defined quantity of tetrahydrofuran to which water was slowly added to prepare spherical NPs.[122]

Lignin nanocarriers were synthesized using a mini-emulsion polymerization method followed by subsequent solvent evaporation;[123] they were proposed as excellent candidates for drug delivery vehicles in the agriculture field or in water purification as carbon materials. Greener lignin NPs as carriers for resveratrol delivery were also synthesized using alkali lignin without any chemical modification employing the self-assembly technique in which water was added to a methanolic solution of alkali lignin; NPs with a perfect spherical shape and better dispersibility ensued. Additionally, the formation of a stable nano-drug carrier was shown by the combination of alkali lignin and Fe_3O_4 magnetic NPs with the bioactive molecule resveratrol where the composition of the solvent plays a significant role in the formation of lignin-based NPs.[107]

A case study on the use of lignin NPs for stimuli-controlled release and entrapment was carried out.[124] The lignin nanoprecipitation (from aqueous ethanol) mechanism and subsequent self-assembly formation were studied in detail, and entrapment of budesonide in lignin NPs was demonstrated. Lignin-based micro- and NPs were also synthesized using a collison-type jet atomizer in which carrier nitrogen gas was used.[125] These particles were then used for the first time to develop coatings on solid supports *via* an evaporation-induced self-assembly approach.[125]

4.2.2. Hydrogels and tissue engineering. Functionalization of lignin NPs with ligands has been reported to load these NPs with an anti-cancer agent and a P-glycoprotein modulator as part of the innovative 4-in-1 strategy.[126] It was concluded from the study that for numerous therapeutic queries, the suggested formulation could be an economical and environmentally friendly remedy. Lignin NPs have also been used in the prepa-

ration of self-healing hydrogels along with polyvinyl alcohol (PVA) and cellulose nanofibrils (CNF) employing dynamic reversible didiol–borax linkages.[127] The main role of lignin in the hydrogel was to act as a spacer and with the rheological analysis, it was confirmed that lignin NPs improved the G' and G'' values of the composite hydrogel because of their anti-aggregation and superelasticity properties. These hydrogels exhibit higher potential for applications in a number of fields including flexible pressure sensors, tissue scaffolds and soft machines;[127] a comprehensive review embodies the applications of lignin-derived scaffolds for possible tissue engineering applications.[128]

Lignin NPs can be used as cross-linking junctions to fabricate a hydrogel with high mechanical characteristics, a polyacrylamide/lignin NP nanocomposite hydrogel. Consequently, the hydrogel showed high compressive and tensile strengths and significant recoverability. The fracture strength of the polyacrylamide/lignin NP hydrogel under compressive stress was on the order of megapascals, which was several orders of magnitude higher than those of the pure polyacrylamide hydrogel. The synergic progressing effect of the nanocomposite network structure and the strong H-bonding between polymer chains endow the hydrogel with a significant mechanism of distributing the applied load.[129]

4.2.3. Antimicrobial effects. Lignin has the ability to form hybrids with nanosilver or nanochitosan to increase their antibacterial activity.[130,131] In one study, lignin NPs have been produced from switchgrass lignin by applying an ultrasound-assisted alkali approach, and were afterwards assembled with gelatin to form a lignin NP–gelatin complex (as a flocculant).[132] After investigation on Gram-positive and Gram-negative strains, the produced complex was found to show relatively better flocculation capacity than lignin NPs, and the flocculation efficiency was significantly influenced by the pH and dosage. For both indicator strains, flocculation efficiency (>95%) was obtained within 30 min at pH 4.5, while a flocculation efficiency of 90% could be reached after 60 min at pH 5; such complexes show promise for flocculating bacteria in wastewater treatment, enrichment and identification of microorganisms.[132] Additionally, PVA/chitosan hydrogels containing 1 and 3 wt% of lignin NPs have been produced *via* the freeze–thaw method.[133] It was illustrated that the strong interaction between PVA/chitosan molecules and lignin NPs prevented the PVA molecules from moving and dissolving in water (Fig. 10 and 11). Findings from microstructural, thermal and mechanical evaluations of the lignin NP-based PVA/chitosan complex showed that the lowest amount of lignin NPs (1 wt%) was useful, whereas the presence of agglomerates at higher lignin NP content limited their effect. Furthermore, swelling behaviour was detected for hydrogels containing lignin NPs with respect to PVA/chitosan, because of the generation of a porous honeycomb-like structure. As a result, a

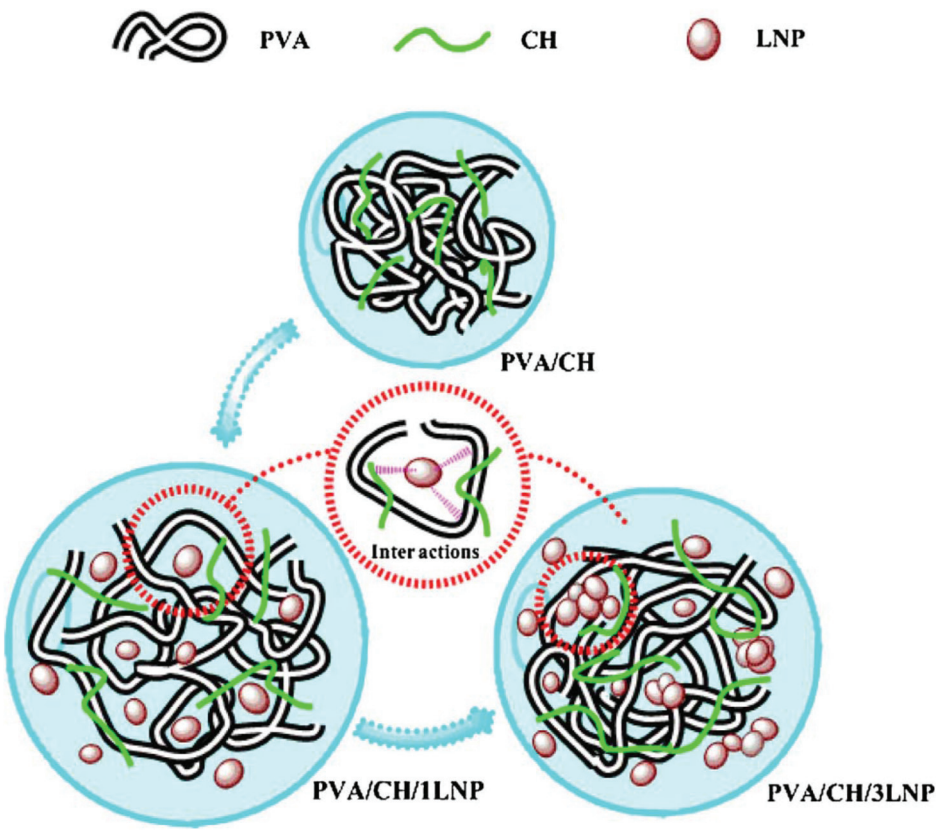


Fig. 10 Interactions between polyvinyl alcohol/chitosan (PVA/Ch) molecules and lignin NPs (LNPs). Reused with permission from ref. 133.

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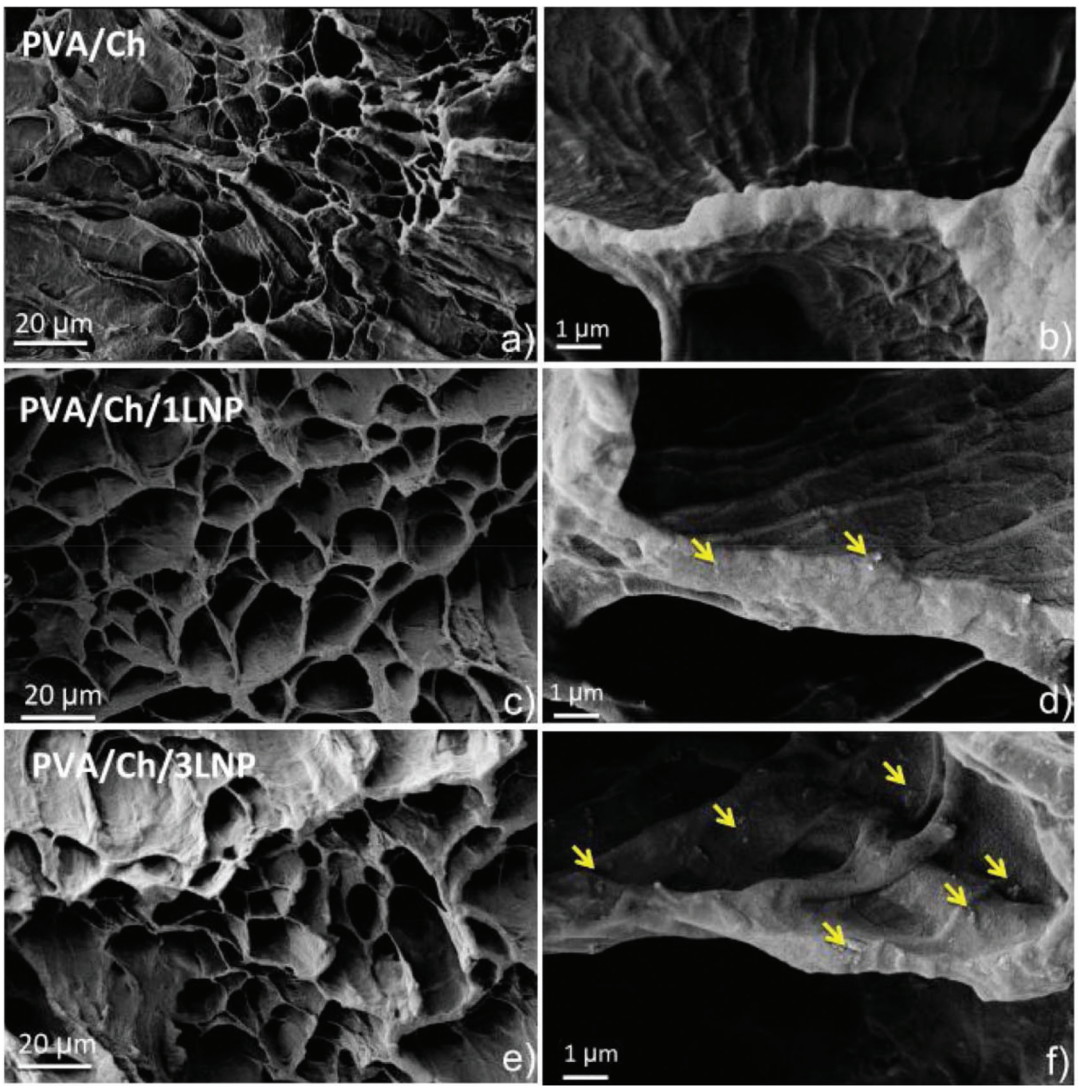


Fig. 11 Morphologies of freeze-dried hydrogels containing different amounts of lignin NPs (PVA/Ch (a), PVA/Ch/1LNP (c) and PVA/Ch/3LNP (e)) were observed, and magnification of lignin NP aggregates (PVA/Ch (b), PVA/Ch/1LNP (d) and PVA/Ch/3LNP (f)) is shown. Polyvinyl alcohol = PVA; chitosan = Ch; and lignin NP = LNP. Reused with permission from ref. 133.

synergetic effect of chitosan and lignin NPs was revealed in terms of the antioxidative response by DPPH (1,1-diphenyl-2-picryl-hydrazyl) activity of migrated substances, whereas findings from antimicrobial evaluations established lignin NPs as effective against Gram negative bacteria when compared to Gram positive ones. The produced hydrogels incorporating lignin NPs can be applied for drug delivery, food packaging, and wound dressing.[133]

Because of the short-term adhesiveness, poor mechanical characteristics and lack of antibacterial ability of traditional adhesive hydrogels, researchers have investigated the plant-inspired adhesive hydrogel which has been prepared based on silver–lignin NP stimulated dynamic redox catechol chemistry (Fig. 12).[134] The silver–lignin NPs construct the dynamic catechol redox system, which produces long-lasting reductive–oxidative environment inner hydrogel networks. This redox system, generating catechol groups continuously, endowed the hydrogel with long-term and repeatable adhesiveness. Additionally, silver–lignin NPs produced free radicals and stimulated self-gelation of the hydrogel in ambient environment. Furthermore, the hydrogel displayed appropriate cell affinity and high antibacterial activity because of the catechol groups and bactericidal ability of silver–lignin NPs (Fig. 13).[134]

4.3. Other miscellaneous applications of lignin NPs

Coir lignin has been used to develop nanocolorants with diverse colours using the electrolysis technique with the variation of the concentration of lignin and the process parameters. These nano-colorants were found to be more fluorescent in nature when compared to the pristine lignin; they can support the cell division with minimum photo-quenching and without any physical damage.[70] Ultrasonic treatment has

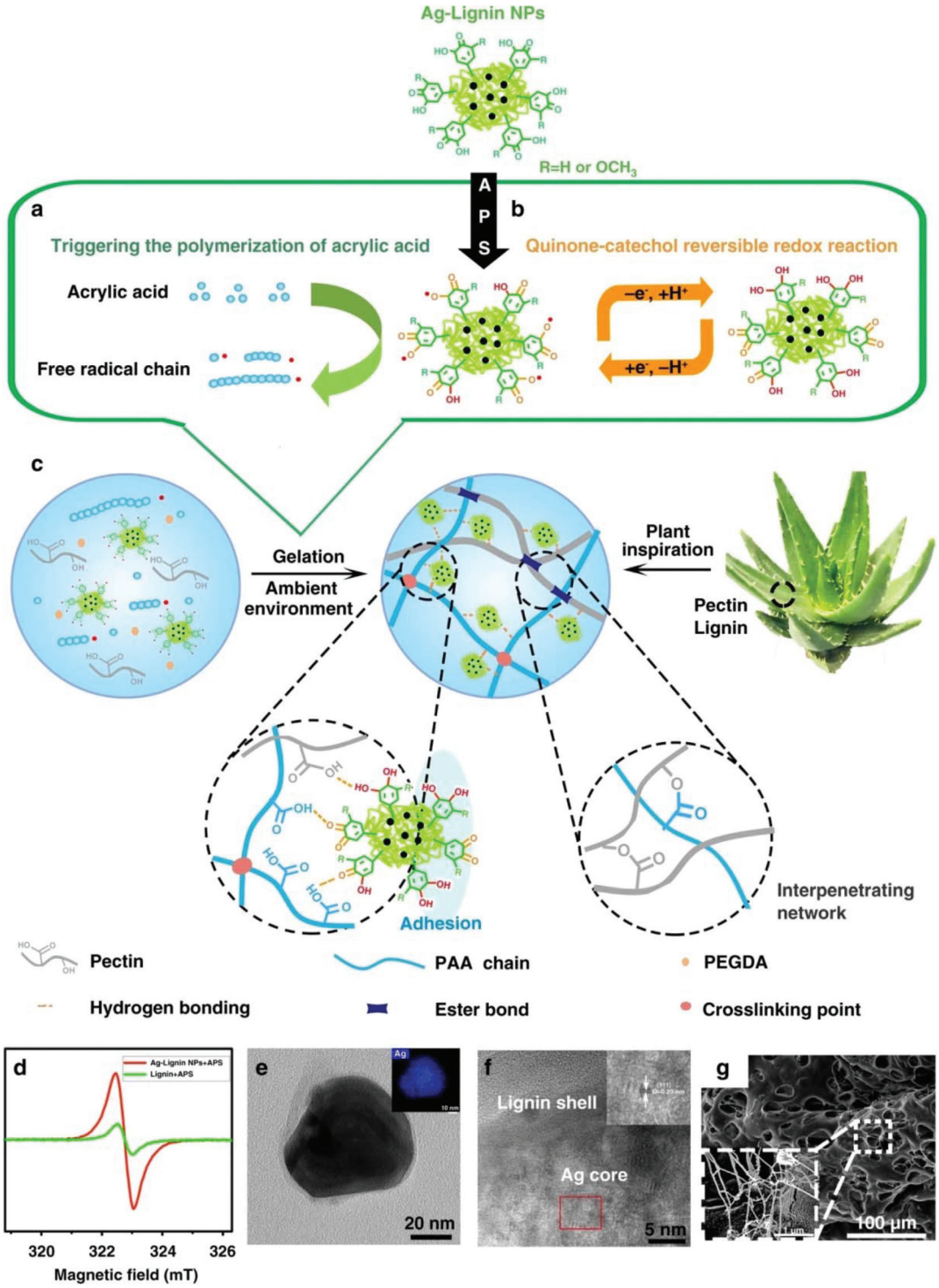


Fig. 12 Design strategy for the plant-inspired catechol-chemistry-based self-adhesive, tough, and antibacterial NP-P-PAA hydrogel. (a) Formation of radicals by the redox reaction between silver–lignin NPs and ammonium persulfate (APS), stimulating the gelation of the hydrogel in an ambient environment. (b) Quinone–catechol reversible reaction maintains a dynamic balance. (c) Scheme of the molecular structure of the plant-inspired adhesive and tough hydrogel. (d) Electron spin resonance spectroscopy (ESR) spectra for quinone radical detection. (e) TEM micrograph demonstrates the core–shell structure of silver–lignin NPs. (f) High-resolution transmission electron microscopy (HRTEM) micrograph demonstrates the silver–lignin NP structure. (g) Scanning electron microscopy (SEM) micrograph demonstrates the microfibril structures in the hydrogel. P, pectin and PAA, polyacrylic acid. Reused from ref. 134, An Open Access Article (CC BY).

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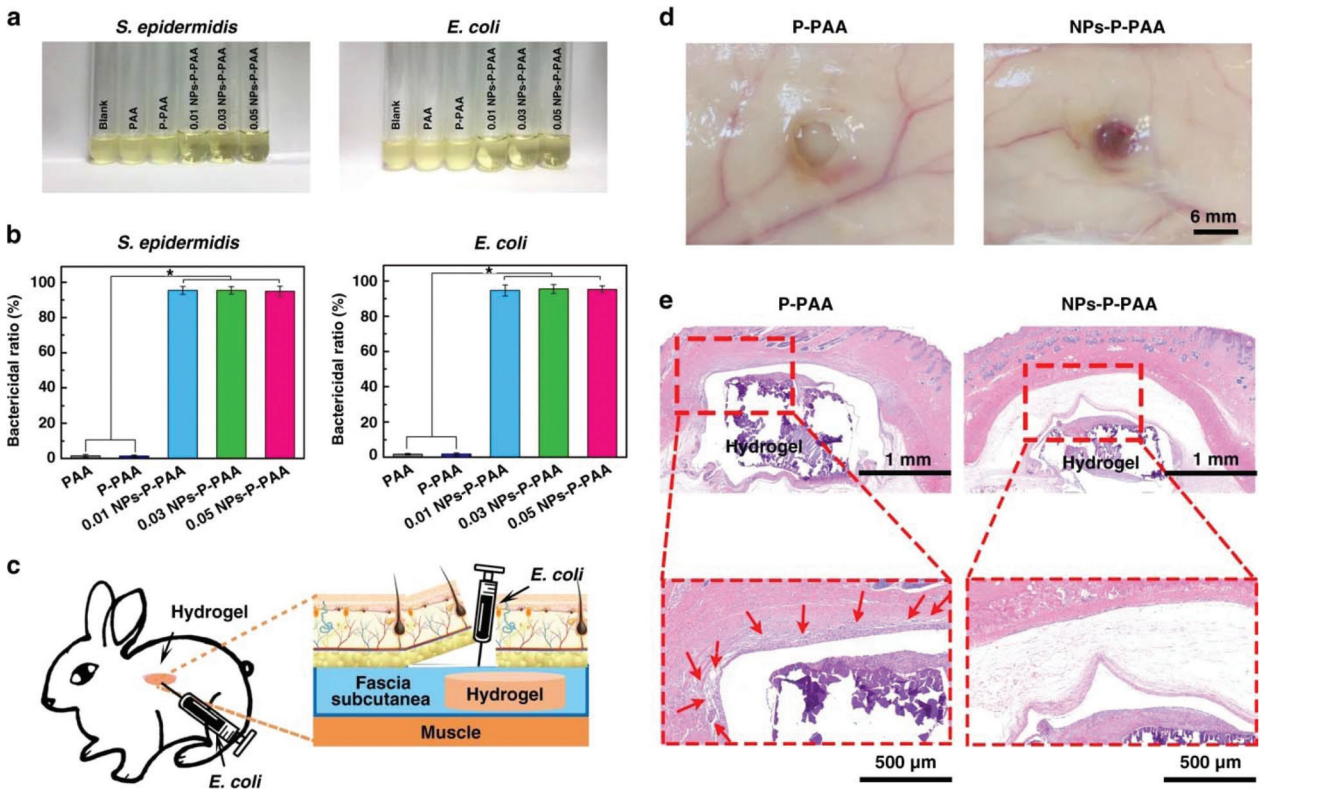


Fig. 13 Antibacterial activities of the hydrogel are reported. (a) Photographs of *S. epidermidis* and *E. coli*. solution co-cultured with the hydrogels after one day. (b) The bactericidal ratio of the hydrogels to *S. epidermidis* and *E. coli*. (c) Scheme of the *in vivo* antibacterial experiments. (d) Photographs of harvested hydrogels after they were implanted in skin pockets for seven days post-surgery. (e) Hematoxylin–eosin-stained sections of connective tissues surrounding the hydrogel. Reused from ref. 134, An Open Access Article (CC BY).

been employed to synthesize lignin NPs from softwood Kraft lignin for subsequent use as lignin–water dispersions.[135] A significant decrease in the particle size of lignin (about 10–50 nm) could be attained *via* tuning the ultrasonic treatment time with the outstanding colloidal stability of the ensuing dispersion with a waterborne thermoplastic polyurethane matrix.

Lignin NPs having a typical size of *ca*. 144 nm were synthesized using a supercritical anti-solvent precipitation method in which acetone was used as a solvent while supercritical carbon dioxide was employed as an anti-solvent.[136] It has been reported that the solubility of nanoscale lignin was 12.4 times greater than that of the parental lignin confirming that the dimensions of lignin exhibit a strong effect on its solubility; the antioxidant activity of the lignin NPs markedly improved as well. Dioxane lignin NPs along with the alkali lignin NPs having a spherical shape with a mean size of 80–104 nm were obtained using a nanoprecipitation method;[137] both the nanomaterials show excellent UV protection and antioxidant properties with potential use in the cosmetic, food and pharmaceutical industries. In another study, lignin NPs have been applied in cosmetic products due to UV absorption.[138]

Composite NPs of lignin/sodium dodecyl sulfate were fabricated for application in Pickering emulsion template-based microencapsulation.[139] Diverse methods namely interfacial crosslinking, anti-solvent precipitation, solvent exchange, polymerization, and sonication deployed for the syntheses and chemical modifications of lignin NPs of various shapes, have been reviewed.[140] Additionally, lignin-based micro- and nanoparticles were synthesized employing ionic liquid (IL) treatment by self-assembly.[141] In this work, wheat grass was used to obtain the alkali lignin which was dissolved using a 1,3-dimethyl-imidazolium dimethyl phosphate IL. Subsequently, water was used to obtain micro/nano-particles of diameters ranging from 200 nm to 1.5 μm *via* precipitation. In another work, nano-lignin as a novel biomaterial to remove red dye was synthesized through acid precipitation technology,[142] particles with diameter of 52.7 nm were obtained at pH = 4 and the adsorption of red dye occurred through pseudo-second order absorption kinetics.[142] Side-streams of pulp/paper along with biorefinery have been deployed to synthesize lignin micro- and NPs; various techno-economic assessments, scalability, and applications have been discussed.[143]

By applying a simple method, lignin NPs of spherical shapes were produced by the application of acids at various pH values (HCl at pH 2.5 and 4.6, H_2SO_4 at pH 2.9 and 4.7 and H_3PO_4 at pH 2.6 and 3.3).[144] In this protocol, alkali lignin was dissolved in ethylene glycol followed by the addition of the acids at several concentrations; lignin NPs were produced in

about 88 wt% yield using acid hydrolysis employing HCl at the pH of 2.5 which was the highest yield obtained among all the acids used. Lignin nanospheres were also synthesized and used for the preparation of transparent UV-absorbing nanocomposite films employing PVA as the polymer matrix;[145] lignin nanospheres were found to be evenly dispersed in the PVA matrix without any macroscopic phase separation when compared to the lignin/PVA blend film. These films are excellent candidates for food packaging and medicine bottle applications. In another study, *N*,*N*-diethylaminoethyl methacrylate (DEAEMA)-grafted lignin NPs were synthesized by using atom transfer radical polymerization, which generated sizes from 237 to 404 nm. These lignin-DEAEMA NPs were used as surfactants for CO_2/N_2-switchable Pickering emulsions correlated with the graft density and chain length of the DEAEMA.[146] Moreover, the anticorrosive behavior of the green synthesized lignin NPs was evaluated, wherein the lignin NPs functioned as anticorrosive nano-fillers for carbon steel protection under stringent corrosive conditions and were dispersed in an epoxy matrix to formulate epoxy nanocomposite coatings. Indeed, NP dispersed epoxy coatings can be used for corrosion protection of the underlying substrate. Various kinds of metal/metal oxide NPs, clay NPs, conducting polymers and graphene have been used for dispersion in epoxy nanocomposite to enhance their anticorrosive behavior and overcome its limitations.[66] Physicomechanical and electrochemical impedance spectroscopy evaluations suggested that lignin NP dispersed epoxy coatings potentially protected the underlying materials and demonstrated better protection as compared to the epoxy coatings themselves.[66]

Lignin can be used as an appropriate support for the immobilization of oxidative enzymes because of its electrochemical responsiveness.[147] For instance, the association between lignin and laccase can enhance the activity of lytic polysaccharide mono-oxygenase in delignification procedures, as a consequence of the generation of low molecular weight lignin derivatives.[148] Additionally, laccase was immobilized on lignin NPs which catalyzed the selective oxidation of alcohols to the corresponding aldehydes by using 2,2,6,6-tetramethylpiperidin-1-yloxyl (TEMPO) as a redox mediator. Lignin was used as an electrochemically active nanostructured support for the immobilization of laccase from *Trametes versicolor* by encapsulation, physical adsorption, and layer by layer technologies, including the glutaraldehyde and bovine serum albumin method. It appears that the accelerated stability and the high activity of the laccase layer by layer functionalized lignin NPs make these systems promising for innovative biotechnological applications.[149] In another study, reactive lignin nanocapsules catalyzed a pigmentation reaction to furnish an innovative type of sustainable polyvalent bioink.[150] In this nanodevice, the pigment, vehicle, binder, and additive were all incorporated into a single confined spherical space. Consequently, bioinks with various shades of colours, black, grey, yellowish, pinkish, and red/brown hues, were produced by choosing the reactants and the pigmentation procedure. It appears that the produced lignin nanocapsules can play multiple roles in the support and activation of the enzyme necessary for producing pigments. Additionally, lignin nanocapsules protected and sheltered the melanin pigment from alkaline and UV-degradation treatment.[150]

Interestingly, it was shown that a mixture of lignin NPs and 1-pentanol produced a biocompatible alternative to non-degradable, synthetic amphiphiles applied for oil spill recovery.[151] The pentanol in the mixture produced initial Marangoni flow and confined the spilled oil into a thick slick on the surface of water. While the alcohol solubilizes, lignin NPs irreversibly adsorb onto the oil–water interface. It was found that the lignin NP adsorption to the oil–water interface was influenced by a combination of electrostatic, van der Waals, and hydrophobic interactions between the particles and the interface. These interactions, combined with interparticle electrostatic repulsion between NPs adsorbed at the oil–water interface, drive the generation of a sub-monolayer. Additionally, the sub-monolayer transformed into a film of jammed NPs because of the compressive stress acting on the interface upon solubilization of pentanol. The interfacial layer of lignin NPs restricted oil from spreading again and locked the oil in its confined state. The herded state of the oil with the interfacial layer of NPs facilitated safe removal of the spilled oil by applying mechanical techniques (Fig. 14).[151]

Lignin NPs (~200 nm) loaded with pyranoanthocyanin dimers (PD) and portisins (P) have been prepared by applying nanoprecipitation and dialysis methods.[152] The stability of lignin NPs was followed at pH ≥5 for at least seven days and the lignin NP encapsulated systems (PD-lignin NPs and P-lignin NPs) demonstrated acceptable color stability towards the pH change after 8 days of incubation, when compared with the pure bluish pigments. It was revealed that lignin NPs could be prepared for the entrapment and stabilization of bluish pyranoanthocyanin pigments through dialysis. It appears that this natural and abundant polymer, obtainable as plentiful waste, from the pulping paper processes can be applied as a tool to stabilize the color of natural bluish dyes and for solubility improvement in aqueous systems for putative food and cosmetic applications.[152] Additionally, a nanoadsorbent has been prepared from Kraft lignin derived from paper and pulp black liquor, chitosan, and titania, and was used for removing the Brilliant Black dye from aqueous solution.[153] The existence of NH, O and TiO functional groups were established by attenuated total reflection (ATR)-Fourier-transform infrared spectroscopy (FTIR), and thermogravimetric evaluation showed that the nanoadsorbent was thermally stable up to 300 °C; the ensuing nanoadsorbent could be effectively used for removing the Brilliant Black dye from contaminated wastewater.[153]

A green and facile production of a versatile lignin–gold NP composite has been reported, which was readily and remotely encapsulated to produce innovative liquid marbles.[154] As a result, the prepared marbles could stay suspended in water, and demonstrated superb photothermal conversion characteristics, as well as visual detection and adsorption towards Pb^{2+}. The marbles could concurrently remotely detect and adsorb

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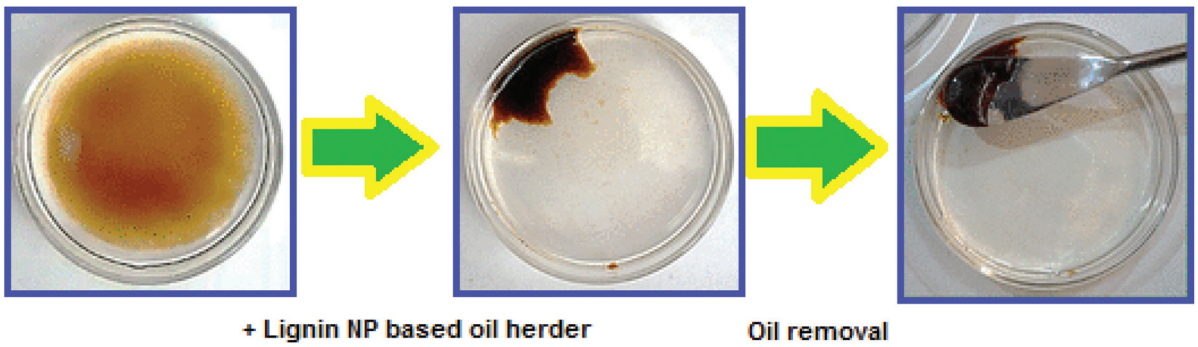


Fig. 14 The transfer of herded oil slick into a fresh water surface; the crude oil herded with a lignin NP–pentanol mixture retained its herded state on the fresh water surface. The capability of retaining the droplet shape highlighted that the lignin NPs were irreversibly adsorbed onto the interface, which further restricted re-spreading of the oil onto the water surface. Adapted with permission from ref. 151. Copyright (2018) American Chemical Society.



Fig. 15 Green preparation of a versatile lignin–gold NP composite. The prepared NIR-responsive lignin–gold NP liquid marbles as remotely controllable miniature reactors for Pb^{2+} detection and adsorption. Reused from ref. 154, An Open Access Article (CC BY).

Pb^{2+} through co-precipitation by applying simple controlled near infrared irradiation (NIR); this remotely controllable NIR-responsive lignin–gold NP liquid marble can be applied in Pb^{2+}-related reactions. This facile manipulation strategy did not use any complicated nanostructures or sophisticated equipment, and thus has potential for applications in channel-free microfluidics, smart microreactors, and microengines (Fig. 15).[154]

5. Challenges and opportunities

Bio-renewable polymers have emerged as an attractive alternative to conventional metallic and organic materials for various applications, because of their biodegradability, biocompatibility and low cost of preparation.[155,156] Renewable resources are garnering increasing attention as a source of environmentally benign biomaterials, such as drug encapsulation/release compounds, scaffolds for tissue engineering, and drug delivery systems for cancers.[73,78,85,147,148,157] It appears that lignin NPs can be produced from different lignin species under various processing pH values at room temperature without any other modifications. Additionally, these NPs can be prepared in high yields, which is promising for large scale production. Lignin NPs offer a way for high-end value utilization of these renewable and abundant resources. However, there were fewer green and facile methods available until now for producing lignin NPs in both higher yields and defined regular shapes. It

appears that the lignin precursor and the NP synthesis methods influence the physical characteristics of the produced NPs; a wide variation of the surface characteristics of lignin NPs makes them promising for various eco-friendly applications.[75] Additionally, important challenges arise from the complexity and heterogeneity of the lignin structures and the resulting molecular properties, the environment, variability of the biomass sources, and pre-treatment procedures. It should be noticed that application of safer, greener and simple approaches for synthesizing lignin NPs with wide aptness and high efficiency will broaden their applications. If the conditions for lignin NP synthesis become so facile, drug-loaded lignin NPs can be produced simultaneously; lignin NPs can also have the ability of higher loading and sustained release of various drugs. Lignin NPs bearing the phenolic OH, COOH and many other functional groups can be simply modified for targeted tumor therapy, making them promising and innovative drug carriers.[73,107,147,148,157]

Indeed, lignin's poor water solubility and its highly branched and random structure make it a challenging biopolymer to exploit when developing novel technologies for producing tailored nanobiomaterials for value-added applications. Various academic investigations focusing on the fabrication and modification of technical lignin in nanoparticulate morphologies demonstrate that these challenges could be overcome by using lignin in the form of colloidal particles. These particles are very stable over a wide pH range and are simply dispersible in organic solvents after stabilization *via* cross-linking. Negative hydroxyl groups on the colloidal particle surface enable multiple enzymatic and chemical modifications, as exemplified through polymerization reactions and surface-coating with positive polymers.[48]

Lignin with its functional groups can be applied as feedstock for an array of organic compounds. Lignin is biodegradable, biocompatible, and antioxidant with no cytotoxicity and its soil nutrient improvement by microbial degradation makes it a suitable precursor for developing eco-friendly nanoscale materials. Lignin is available in the form of industrial/agricultural biomass as a waste, and is among the low-cost biomaterials, with the main cost being predominantly its transportation to a processing facility.[158] Biomass and industrial waste as feedstocks for diverse synthesis uses will help promote them for the sustainability of the environment. Production of lignin NPs and their numerous high-tech applications are the ideal pathways for applying and reducing biowaste from agriculture and industries.[122] Many synthetic methods have been reported for producing lignin NPs involving hazardous materials (*e.g.*, solvents such as THF) but it is vital, especially for industrial production, to reduce or even eliminate their application or to substitute them with greener solvents.[159] It is essential to develop innovative methods for well-organized, inexpensive, time-saving, and cost-effective production of lignin NPs under eco-friendly conditions with the desired structures, properties, and morphologies appropriate for applications, especially in the highly valued field of nano- and bio-medicine.[159]

Despite the large-scale production of lignin, its utilization in high-value applications has remained an important challenge. Bountiful explorations on lignin-derived composites do confirm its potential for scaffolding and/or drug release development. In academic investigations, lignin NPs demonstrated potential uses in composites, UV absorbents and antioxidant formulations, and the possibility of using lignin NPs in high-value medical applications with appealing results. Although the isolation of well-defined lignin fractions is still demanding, the intrinsic bioactivity will be the driving force for the fruitful application of lignin-derived biomaterials in medicine. Additionally, one of the important defying aspects is the reactivity of lignin that is significantly limited because of the high degree of substitution in the *para* position of the aliphatic chain and in the *ortho* position by the methoxyl group culminating in steric hindrance.[160] It appears that large-scale fabrication of lignin NPs can be taxing as the process requires handling of solvents. Furthermore, the challenge of utilization of Lignin NPs lies in their applicable incorporation into polymer matrices. Innovative synthetic methods, including crosslinking and polymer grafting, have been suggested for producing lignin NP-based products. In addition, the undesirable color of lignin would be an important concern for its utilization in certain applications, such as cosmetic products.

6. Conclusion

Lignin, an environmentally sustainable byproduct of the pulp and paper industries, contains a multitude of phenolic hydroxyl groups, some of which are known to readily and strongly chelate with ions. The shape, size, and stability of LNPs are dependent on the lignin type, preparation processes, and production conditions. Currently, elaborative efforts are underway in research laboratories and industries with a focus on lignin applications as renewable macromolecular building blocks for the production of polymeric drug encapsulation and scaffold materials. Lignin NPs, as renewable green materials, can be promising candidates for various biomedical applications, including drug delivery, cancer therapy and diagnosis. The antioxidant and antimicrobial capacity of lignin extracts permits a broad variety of potential applications, such as drugs in cancer therapy. Additionally, lignin NPs show great potential to be applied as suitable oral drug delivery systems, especially for poorly soluble drugs with limited bioavailability. Rather than treating lignin as waste, the fabrication of lignin NPs and polymers has attracted attention for inducing lignin-based value-added products. However, it should be mentioned that comprehensive and systematic scale-up evaluations and analyses of processes for producing lignin NPs and polymers at pilot and pre-commercial scales are essential for undertaking the important challenges which may be associated with the large-scale commercialization of these products. Importantly, the problems encountered in the preparation of value-added products from lignin are correlated with its heterogeneity at a molecular level and limited reactivity as well as

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differences in the characteristics of lignin associated with lignin's origin and its extraction procedures.

Conflicts of interest

The authors declare no conflict of interest.

Acknowledgements

The work was supported by the ERDF project "Development of pre-applied research in nanotechnology and biotechnology" (No. CZ.02.1.01/0.0/0.0/17_048/0007323).

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